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5 **Draft Standard for**
6 **Local and metropolitan area networks—**
7 **Timing and Synchronization for**
8 **Time-Sensitive Applications**

9 **Amendment: Support for the IEEE Std 802.3**
10 **Clause 4 Media Access Control (MAC) operating in**
11 **half-duplex**

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16 **Time-Sensitive Networking (TSN) Task Group of IEEE 802.1**

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1 Editors' Foreword

2 This draft standard is an amendment. The scope of changes to the base standard is thus strictly limited, as
3 detailed in the [PAR](#).

4 Information on participation in this project, and in the IEEE 802.1 Working Group can be found [here](#).

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23 802.1 Working Group.

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33 referred to as the 5 Criteria or 5C's) is reviewed on a regular basis in order to ensure their continued validity.
34 A vote of "Approve" on this draft is also an affirmation by the balloter that the PAR is still valid.

1 Project Authorization Request, Scope, Purpose, and Criteria for Standards 2 Development (CSD)

3 The complete amendment PAR, as approved by IEEE NesCom 23 February 2022, can be found at:

4 <https://development.standards.ieee.org/myproject-web/public/view.html#pardetail/9522>

5 The 'Scope of the Proposed changes' and the 'Need for the Project' specify the changes to be made by this
6 amendment (see below).

7 Scope of the Proposed changes:

8 This amendment specifies protocols, procedures, and managed objects that support IEEE Std 802.3 Clause 4
9 Media Access Control (MAC) operating in half-duplex while retaining existing functionality and backward
10 compatibility, and remaining a profile of IEEE Std 1588™-2019.

11 This amendment addresses errors and omissions in the description of existing functionality.

12 Need for the Project:

13 Support is needed in applications such as automotive in-vehicle networks and industrial automation networks
14 for the IEEE Std 802.3 Clause 4 MAC operating in half-duplex, including those using links with the
15 10BASE-T1S PHY in either point-to-point or multidrop half-duplex mode recently introduced by IEEE Std
16 802.3cg-2019.

17 Criteria for Standards Development:

18 The complete Criteria for Standards Development (CSD) can be found at:

19 <https://mentor.ieee.org/802-ec/dcn/21/ec-21-0308-00-ACSD-p802-1asds.pdf>

20

**Draft IEEE Standard for
Local and metropolitan area networks—**

**Timing and Synchronization for
Time-Sensitive Applications**

**Amendment: Support for the IEEE Std 802.3
Clause 4 Media Access Control (MAC) operating in
half-duplex**

Prepared by the
Time-Sensitive Networking (TSN) Task Group of IEEE 802.1

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1 **Abstract:** This amendment to IEEE Std 802.1AS™-2020 specifies protocols, procedures, and
2 managed objects that support IEEE Std 802.3 Clause 4 Media Access Control (MAC) operating in
3 half-duplex while retaining existing functionality and backward compatibility, and remaining a profile
4 of IEEE Std 1588™-2019.

5 This amendment addresses errors and omissions in the description of existing functionality.

6 **Keywords:** best timeTransmitter, frequency offset, Grandmaster Clock, Grandmaster PTP
7 Instance, PTP End Instance, PTP Relay Instance, IEEE Std 802.1AS™, phase offset,
8 synchronization, syntonization, time-aware system

The Institute of Electrical and Electronics Engineers, Inc.
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PDF:	ISBN 978-X-XXX-XXX-X	STDXXXXX
Print:	ISBN 978-X-XXX-XXX-X	STDPDXXXXX

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6 **János Farkas, *TSN Task Group Chair***
7 **Silvana Rodrigues, *Editor IEEE Std 802.1AS***
8 **Silvana Rodrigues, *Editor P802.1ASds***
9

10 The following members of the individual balloting committee voted on this standard. Balloters may have
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10

1 Introduction

This introduction is not part of IEEE Std 802.1ASdsTM-20xx, IEEE Standard for Local and metropolitan area networks—Timing and Synchronization for Time-Sensitive Applications—Amendment: Support for the IEEE Std 802.3 Clause 4 Media Access Control (MAC) operating in half-duplex

2 The first edition of IEEE Std 802.1AS was published in 2011. A first corrigendum, IEEE Std
3 802.1ASTM-2011/Cor1-2013, provided technical and editorial corrections. A second corrigendum, IEEE Std
4 802.1ASTM-2011/Cor2-2015 provided additional technical and editorial corrections.

5 The second edition, IEEE Std 802.1AS-2020, added support for multiple gPTP domains, Common Mean
6 Link Delay Service, external port configuration, and Fine Timing Measurement for 802.11 transport.
7 Backward compatibility with IEEE Std 802.1AS-2011 was maintained. A corrigendum, IEEE Std
8 802.1ASTM-2020/Cor1-2021, provides technical and editorial corrections.

9 The third edition, IEEE Std 802.1AS-202x is a roll-up of IEEE Std 802.1AS-2020 with the corrigendum
10 IEEE Std 802.1AS-2020/Cor1, and its amendments: IEEE Std 802.1ASdr, IEEE Std 802.1ASdn, and IEEE
11 Std 802.1ASdm.

12 This amendment to IEEE Std 802.1AS-202x specifies protocols, procedures, and managed objects that
13 support IEEE Std 802.3 Clause 4 Media Access Control (MAC) operating in half-duplex while retaining
14 existing functionality and backward compatibility, and remaining a profile of IEEE Std 1588TM-2019.

15 This amendment addresses errors and omissions in the description of existing functionality

16 <<Editor's note: P802.1ASds is an amendment to 802.1AS-2025 that is the outcome of
17 P802.1AS-2020-REV.>>

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1

2 **Draft IEEE Standard for** 3 **Local and metropolitan area networks—** 4 **Timing and Synchronization for Time-** 5 **Sensitive Applications** 6 **Amendment: Support for the IEEE Std 802.3** 7 **Clause 4 Media Access Control (MAC) operating in** 8 **half-duplex**

9 [This amendment is based on IEEE Std 802.1ASTM-20xx (IEEE Std 802.1ASTM-2020 Revision).

10 NOTE—The editing instructions contained in this amendment define how to merge the material contained therein into
11 the existing base standard and its amendments to form the comprehensive standard.

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19 changes will be incorporated into the base standard.¹

20

¹Notes in text, tables, and figures are given for information only, and do not contain requirements needed to implement the standard.

13. Definitions

Insert the following definitions in Clause 3, and renumber the definitions as appropriate:

3.17 Half-duplex Ethernet: An IEEE Std 802.3 (Ethernet) link whose physical ports use the IEEE Std 802.3TM Clause 4 MAC in half-duplex mode, independent of the physical medium access method and independent of the physical topology of the medium connected (which is typically a shared medium).

4. Acronyms and abbreviations

Insert the following acronym in clause 4 as follows:

HDE Half-duplex Ethernet

4

5. Conformance

Change the title of 5.5 as follows:

5.5 .MAC-specific timing and synchronization methods for full-duplex IEEE Std 802.3-2022 links

Change the first and sentence of 5.5 as follows:

An implementation of a time-aware system with full-duplex IEEE Std 802.3-2022 media access control (MAC) services to physical ports shall:

Change the sentence after item e) of 5.5 as follows:

An implementation of a PTP Instance with full-duplex IEEE Std 802.3-2022 MAC services to physical ports may:

Insert 5.9 as follows:

5.9 MAC-specific timing and synchronization methods for half-duplex Ethernet (HDE) links

An implementation of a time-aware system with half-duplex IEEE Std 802.3-2022 media access control (MAC) services to physical ports shall:

- a) Support half-duplex operation, as specified in Clause 4 of IEEE Std 802.3-2022.
- b) Support the requirements as specified in Clause 19.

18

17. Time-synchronization model for a packet network

2 7.2 Architecture of a time-aware network

3 7.2.1 General

4 *Add item g) after item f) in the lettered list in 7.2.1 as follows:*

- 5 g) IEEE Std 802.3 Clause 4 Media Access Control (MAC) operating in half-duplex as specified in
6 Clause 19.

7 7.6 Differences between gPTP (IEEE Std 802.1AS) and PTP (IEEE Std 1588-2019)

8 *Change item e) and item f) in 7.5 as follows:*

- 9 e) For full-duplex Ethernet links and HDE links, gPTP requires the use of the peer-to-peer delay
10 mechanism, while IEEE Std 1588-2019 also allows the use of end-to-end delay measurement.
- 11 f) For full-duplex Ethernet links and HDE links, gPTP requires the use of two-step processing (use of
12 Follow_Up and Pdelay_Resp_Follow_Up messages to communicate timestamps), with an optional
13 one-step processing mode for full-duplex Ethernet that embeds timestamps in the Sync “on the fly”
14 as they are being transmitted (gPTP does not specify one-step processing for peer delay messages).
15 IEEE Std 1588-2019 allows either two-step or one-step processing to be required (for both Sync and
16 peer delay messages) depending on a specific profile.

1 8. IEEE 802.1AS concepts and terminology

2 8.5 Ports

3 8.5.1 General

4 *Change 8.5.1 as follows:*

5 The PTP Instances in a gPTP domain interface with the network media via physical ports. gPTP defines a
6 ~~logical port, i.e.,~~ a PTP Port, in such a way that communication between PTP Instances is point-to-point or,
7 point-to-multipoint. The choice of point-to-point (e.g., see Clause 16) or point-to-multipoint (e.g., see
8 Clause 19) communication is specified in each media dependent specification in this standard. even over
9 ~~physical ports that are attached to shared media. One logical~~ A PTP port, consisting of one PortSync entity
10 and one media-dependent (MD) entity, ~~is instantiated for each PTP Instance with which the PTP Instance~~
11 ~~communicates. For shared media, multiple logical ports can be associated with a single physical port.~~
12 ~~Unless otherwise qualified, each instance of the term port refers to a logical port.~~

11. Media-dependent layer specification for full-duplex point-to-point links

11.1 Overview

11.1.1 General

Add a NOTE at the end of 11.1.1 as follows:

NOTE—PTP links using the IEEE Std 802.3 Clause 4 MAC operating in half-duplex mode are specified in Clause 19.

11.2 State machines for MD entity specific to full-duplex point-to-point links

11.2.1 General

Replace Figure 11-5 with the following:

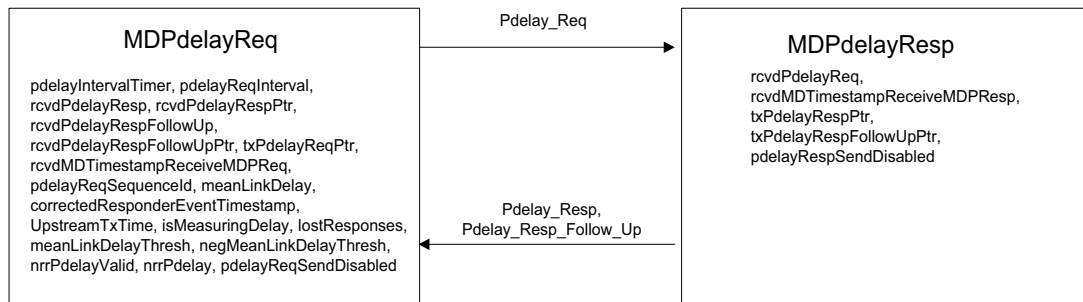


Figure 11-5—Peer-to-peer delay mechanism state machines—overview and interrelationships

11.2.2 Determination of asCapable and asCapableAcrossDomains

Change 11.2.2 as follows:

There is one instance of the global variable asCapable (see 10.2.5.1) per PTP Port, per domain. For full-duplex point-to-point links, there is one instance of the global variable asCapableAcrossDomains (see 11.2.13.13), per port, that is common across, and accessible by, all the domains. For HDE links, there is one instance of asCapableAcrossDomains per PTP Instance (i.e., per domain) (see 11.2.13.13).

The ~~per-PTP Port~~ global variable asCapable (see 10.2.5.1) indicates whether the IEEE 802.1AS protocol is operating, in this domain, on the PTP Link attached to this PTP Port, and can provide the required time-synchronization performance ~~described in B.3~~. asCapable is used by the PortSync entity, which is media-independent; however, the determination of asCapable is media-dependent.

The per-port global variable asCapableAcrossDomains is set by the MDPdelayReq state machine (see 11.2.19 and Figure 11-9). For a port attached to a full-duplex point-to-point PTP Link or to an HDE link, asCapableAcrossDomains shall be set to TRUE if and only if either:

- 1 a) ~~It~~ is determined, via the transport-specific peer-to-peer delay mechanism or CMLDS, that the
 2 following conditions hold for the port:
- 3 1) ~~⊕~~The port is exchanging peer delay messages with its neighbor,
 4 2) ~~⊕~~The measured delay does not exceed meanLinkDelayThresh__or is below
 5 negMeanLinkDelayThresh,
 6 3) ~~⊕~~For full-cuplex links, ~~⊕~~the port does not receive multiple Pdelay_Resp or
 7 Pdelay_Resp_Follow_Up messages in response to a single Pdelay_Req message, ~~and~~
 8 4) For HDE links, the port does not receive multiple Pdelay_Resp or multiple
 9 Pdelay_Resp_Follow_Up messages with the requestingPortIdentity.Clock Identity equal to the
 10 PTP Instance's clock identity in response to a single Pdelay_Req message, and
 11 5) ~~⊕~~The port does not receive a response from itself or another PTP Port of the same PTP
 12 Instance.

13 or:

- 14 b) pdelayReqSendDisabled is set to TRUE.

15 NOTE 1—If a PTP Instance implements only domain 0 and the MDPdelayReq and MDPdelayResp state machines are
 16 invoked on domain 0 (see 11.2.19), asCapableAcrossDomains is still set by the MDPdelayReq state machine.

17 The default value of meanLinkDelayThresh shall be set as specified in Table 11-1.

Table 11-1—Value of meanLinkDelayThresh for various links

Link	Value of meanLinkDelayThresh (ns) (see NOTE)
100BASE-TX, 1000BASE-T	800 ₁₀
100BASE-FX, 1000BASE-X	0x7FFFFFFF
NOTE—The actual propagation delay for 100BASE-TX and 1000BASE-T links is expected to be smaller than the above respective threshold. If the measured mean propagation delay (i.e., meanLinkDelay; see 10.2.5.8) exceeds this threshold, it is assumed that this is due to the presence of equipment that does not implement gPTP. For 100BASE-FX and 1000BASE-Xlinks, the actual propagation delay can be on the order of, or larger than, the delay produced by equipment that does not implement gPTP; therefore, such equipment cannot be detected by comparing measured propagation delay with a threshold. In this case, meanLinkDelayThresh is set to the largest possible positive value (i.e., in binary notation a zero followed by all 1s). The default value for negMeanLinkDelayThresh is not defined and it is implementation specific. <u>This table does not list propagation delay threshold values for all types of PHYs because the values are implementation specific and depend on the application.</u>	

18 The per-PTP Port, per-domain global variable asCapable shall be set to TRUE if and only if the following
 19 conditions hold:

- 20 c) ~~⊕~~The value of asCapableAcrossDomains is TRUE, and
 21 d) ~~⊕~~One of the following conditions holds:
- 22 1) The value of neighborGptpCapable for this PTP Port is TRUE, or
 23 2) The value of domainNumber is zero, and the value of sdoId for peer delay messages received
 24 on this PTP Port is 0x100.

25 NOTE 2—Condition ~~⊕~~d) 2) ensures backward compatibility with the 2011 edition of this standard. A PTP Instance
 26 compliant with the current edition of this standard that is attached, via a full-duplex point-to-point PTP Link, to a PTP

1 Instance compliant with the 2011 edition of this standard will not receive Signaling messages that contain the gPTP
2 capable TLV and will not set neighborGtpCapable to TRUE. However, condition ~~d~~ 2) ensures that asCapable for
3 this PTP Port and domain (i.e., domain 0) will still be set to TRUE if condition c) holds because the peer delay messages
4 received from the time-aware system compliant with the 2011 edition of this standard will have sdold set to 0x100.

5 11.2.12 Overview of MD entity global variables

6 *Add the following note in Table 11-2:*

7 NOTE — asCapableAcrossDomains is per PTP Instance for HDE links.

8 11.2.13 MD entity global variables

9 *Change 11.2.13.13 as follows:*

10 **11.2.13.13 asCapableAcrossDomains:** A Boolean that is TRUE if and only if either: 11.2.2 item a) or
11 11.2.2 item b) is satisfied~~conditions a) through d) of 11.2.2 are satisfied. This Boolean~~
12 asCapableAcrossDomains is set by the MDPdelayReq state machine and is used in determining asCapable
13 for a port (see 11.2.2). There is one instance of this variable for all the domains (per port) for full-duplex
14 point-to-point links.~~And~~ the variable is accessible by all the domains. There is one instance of this variable
15 per PTP Instance (i.e., per domain) for HDE links (see 19.2.2). When only one domain is active,
16 asCapableAcrossDomains is equivalent to the variable asCapable (see 10.2.5.1).

17 11.2.19 MDPdelayReq state machine

18 11.2.19.2 State machine variables

19 *Change 11.2.19.2.2 as follows:*

20 11.2.19.2.2 rcvdPdelayResp:

- 21 a) For full-duplex point-to-point mode: A Boolean variable that notifies the current state machine
22 when a Pdelay_Resp message is received. This variable is reset by the current state machine.
- 23 b) For half-duplex mode: A Boolean variable that notifies the current state machine when a
24 Pdelay_Resp message is received AND its requestingPortIdentity.clockIdentity is equal to the
25 clockIdentity of the current PTP Instance. This variable is reset by the current state machine.

26 *Change 11.2.19.2.4 as follows:*

27 11.2.19.2.4 rcvdPdelayRespFollowUp:

- 28 a) For full-duplex point-to-point mode: A Boolean variable that notifies the current state machine
29 when a Pdelay_Resp_Follow_Up message is received. This variable is reset by the current state
30 machine.
- 31 b) For half-duplex mode: A Boolean variable that notifies the current state machine when a
32 Pdelay_Resp_Follow_Up message is received AND its requestingPortIdentity.clockIdentity is equal
33 to the clockIdentity of the current PTP instance. This variable is reset by the current state machine.

34 *Insert a new variable after 11.2.19.2.13 as follows:*

35 **11.2.19.2.14 pdelayReqSendDisabled:** A boolean that is administratively set to TRUE if Pdelay_Req
36 messages are not transmitted by this port. The default value for this variable shall be FALSE.

11.2.19.4 State diagram

2 Replace Figure 11-9 with the following:.

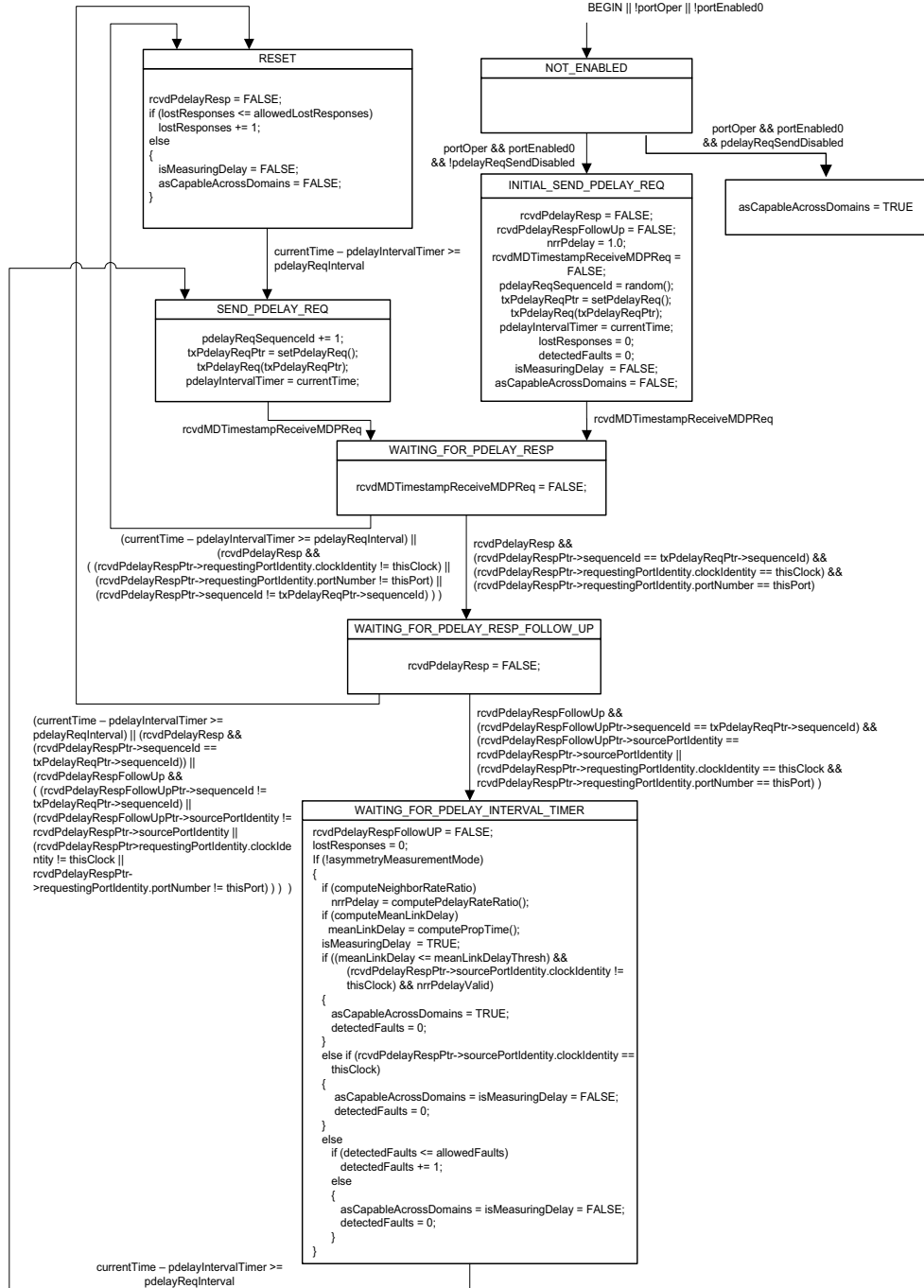


Figure 11-9—MDPdelayReq state machine

1 **Add a NOTE after Figure 11-9 as follows:**

2 NOTE—A change in the value of the variable pdelayReqSendDisabled takes effect only when portEnabled0 (see
 3 11.2.19.2.12) is FALSE.

4 **11.2.20 MDPdelayResp state machine**

5 **11.2.20.2 State machine variables**

6 **Insert a new variable after 11.2.20.2.5 as follows:**

7 **11.2.20.2.6 pdelayRespSendDisabled:** A boolean that is administratively set to TRUE to indicate that
 8 corresponding Pdelay_Resp and Pdelay_Resp_Follow_Up messages are not transmitted by this port in
 9 response to a Pdelay_Req message. The default value for this variable shall be FALSE.

10 **11.2.20.4 State diagram**

11 **Replace Figure 11-10 with the following:**

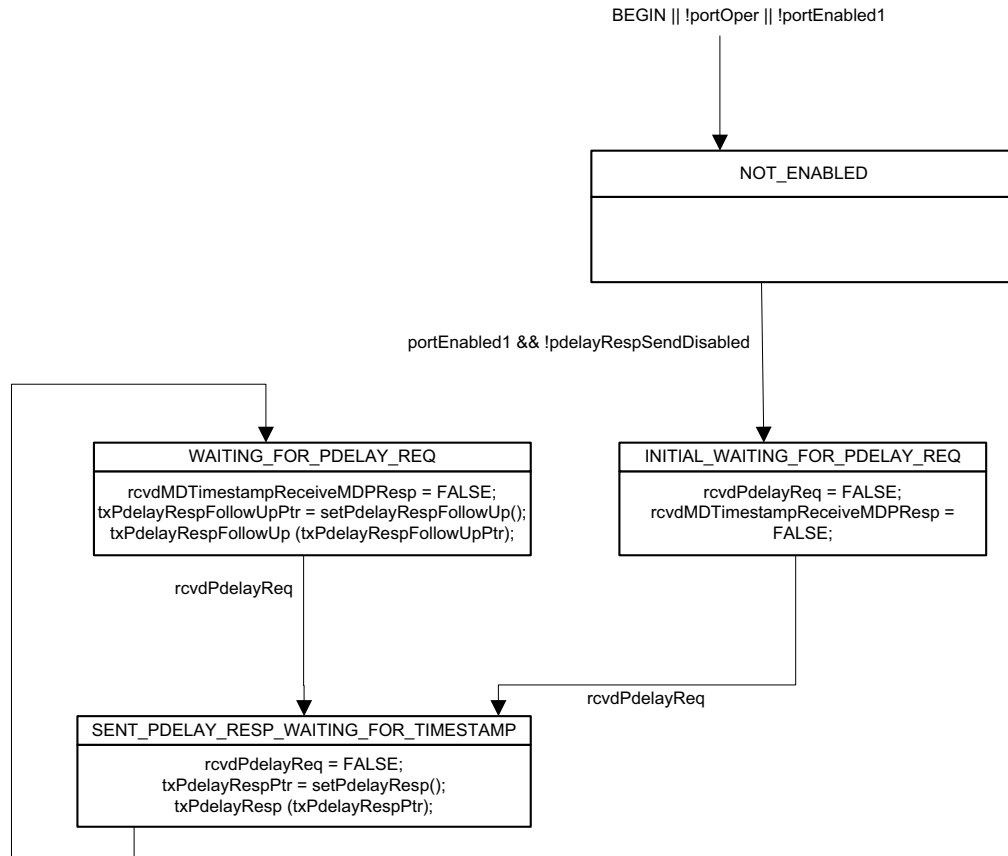


Figure 11-10—MDPdelayResp state machine

12 **Add a NOTE after Figure 11-10 as follows:**

¹NOTE—A change in the value of the variable pdelayRespSendDisabled takes effect only when portEnabled1 (see
² 11.2.20.2.5) is FALSE.

1 14. Timing and synchronization management

2 14.10 Port Parameter Data Set (portDS)

3 14.10.1 General

4 *Insert the following items after the final item of Table 14-9:*

Table 14-9—portDS table

Name	Data type	Operations supported ^a	References
pdelayReqSendDisabled	Boolean	RW	14.10.60
pdelayRespSendDisabled	Boolean	RW	14.10.61

^a R = Read only access; RW = Read/write access.

5 14.10.23 InitialLogPdelayReqInterval

6 *Change the first paragraph in 14.10.23 as follows:*

7 For full-duplex [and half-duplex](#) IEEE [Std](#) 802.3 media and for CSN media that use the peer-to-peer delay
 8 mechanism to measure path delay (see 16.4.3.2), the value is the logarithm to base 2 of the Pdelay_Req
 9 message transmission interval used when:

10 14.10.24 currentLogPdelayReqInterval

11 *Change the first paragraph in 14.10.24 as follows:*

12 For full-duplex [and half-duplex](#) IEEE [Std](#) 802.3 media and for CSN media that use the peer-to-peer delay
 13 mechanism to measure path delay (see 16.4.3.2), the value is the logarithm to the base 2 of the current
 14 Pdelay_Req message transmission interval (see 11.5.2.2).

15 14.10.32 currentComputeNeighborRateRatio

16 *Change the first paragraph in 14.10.32 as follows:*

17 For full-duplex [and half-duplex](#) IEEE [Std](#) 802.3 media and for CSN media that use the peer-to-peer delay
 18 mechanism to measure path delay (see 16.4.3.2), the value is the current value of
 19 computeNeighborRateRatio.

20 14.10.36 currentComputeMeanLinkDelay

21 *Change the first paragraph in 14.10.36 as follows:*

22 For full-duplex [and half-duplex](#) IEEE [Std](#) 802.3 media and for CSN media that use the peer-to-peer delay
 23 mechanism to measure path delay (see 16.4.3.2), the value is the current value of computeMeanLinkDelay.

1 14.10.53 pdelayTruncatedTimestampsArray**2 *Change the first paragraph in 14.10.53 as follows:***

3 For full-duplex and half-duplex IEEE Std 802.3 media and for CSN media that use the peer-to-peer delay
4 mechanism to measure path delay (see 16.4.3.2), the values of the four elements of this array are as
5 described in Table 14-9. For all other media, the values are zero. Array elements 0, 1, 2, and 3 correspond to
6 the timestamps t1, t2, t3, and t4, respectively, in Figure 11-1 and are expressed in units of 2–16 ns (i.e., the
7 value of each array element is equal to the remainder obtained upon dividing the respective timestamp,
8 expressed in units of 2–16 ns, by 248). At any given time, the timestamp values stored in the array are for the
9 same, and most recently completed, peer delay message exchange.

10 *Insert 14.10.60 and 14.10.61 as follows:***11 14.10.60 pdelayReqSendDisabled**

12 The value is equal to the value of the per-PTP Port variable pdelayReqSendDisabled (see 11.2.19.2.14). If its
13 value is TRUE, Pdelay_Req messages are not transmitted by the PTP Port. The default value for this
14 variable shall be FALSE.

15 14.10.61 pdelayRespSendDisabled

16 The value is equal to the value of the per-PTP Port variable pdelayRespSendDisabled (see 11.2.20.2.6). If its
17 value is TRUE, Pdelay_Resp and Pdelay_Resp_Follow_Up messages are not transmitted by the PTP Port in
18 response to Pdelay_Req message. The default value for this variable shall be FALSE.

19 14.10.55 portDS table

20

17. YANG Data Model

17.2 IEEE 802.1AS YANG models

3 Replace Figure 17-3 with the following:

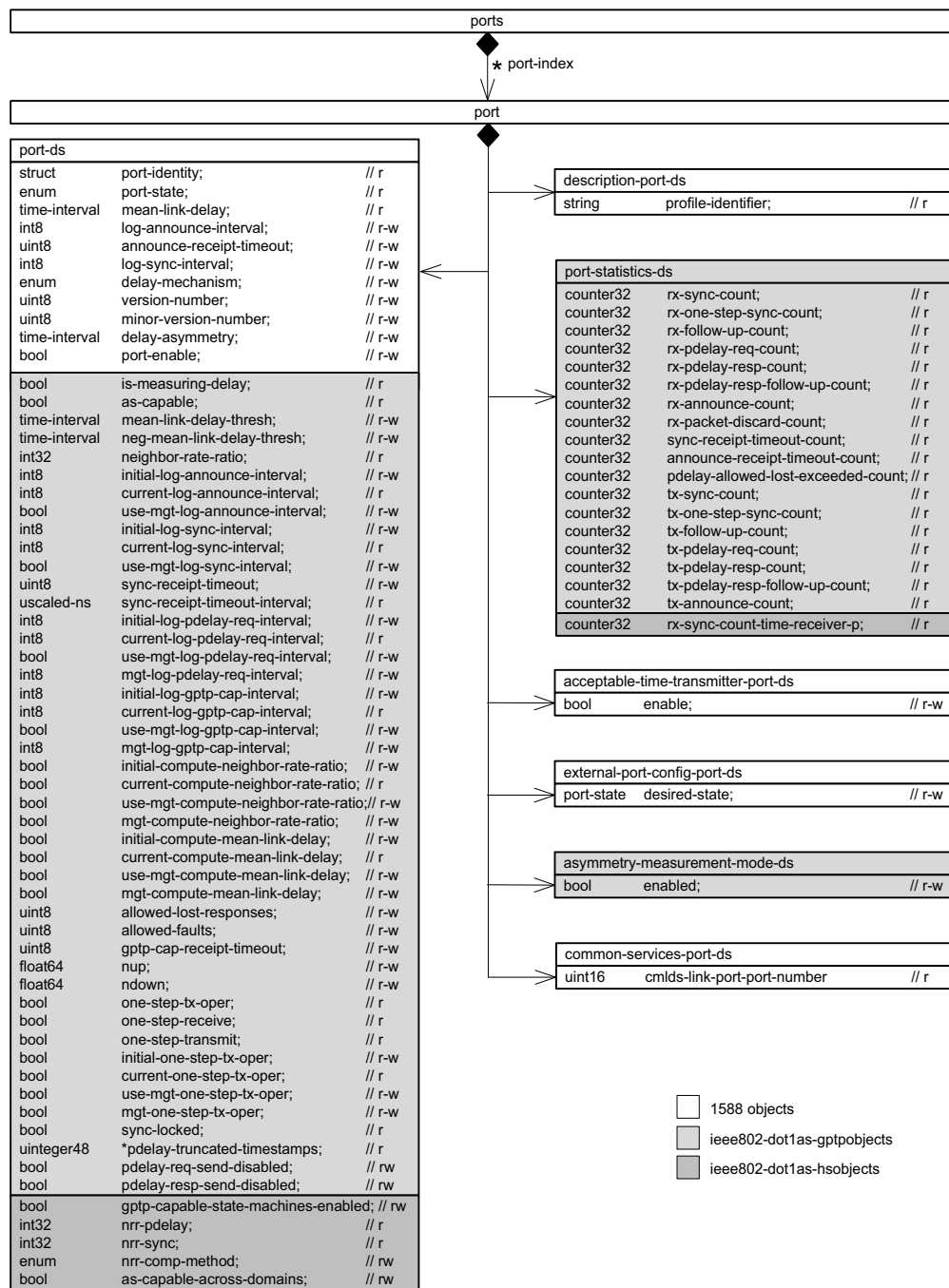


Figure 17-3—PTP Port detail

1 17.5 YANG schema tree definitions

2 17.5.1 ITree diagram for ieee802-dot1as-gptp.yang

3 *Replace the tree diagram for ieee802-dot1as-gptp.yang as follows:*

```

4 module: ieee802-dot1as-gptp
5
6   augment /ptp-tt:ptp/ptp-tt:instances/ptp-tt:instance/ptp-tt:default-ds:
7     +--ro gm-capable?                boolean
8     +--ro current-utc-offset?        int16
9     +--ro current-utc-offset-valid?  boolean
10    +--ro leap59?                    boolean
11    +--ro leap61?                    boolean
12    +--ro time-traceable?             boolean
13    +--ro frequency-traceable?        boolean
14    +--ro ptp-timescale?              boolean
15    +--ro time-source?                identityref
16  augment /ptp-tt:ptp/ptp-tt:instances/ptp-tt:instance/ptp-tt:current-ds:
17    +--ro last-gm-phase-change?       scaled-ns
18    +--ro last-gm-freq-change?        float64
19    +--ro gm-timebase-indicator?      uint16
20    +--ro gm-change-count?            yang:counter32
21    +--ro time-of-last-gm-change?     yang:timestamp
22    +--ro time-of-last-phase-change?  yang:timestamp
23    +--ro time-of-last-freq-change?   yang:timestamp
24  augment /ptp-tt:ptp/ptp-tt:instances/ptp-tt:instance/ptp-tt:parent-ds:
25    +--ro cumulative-rate-ratio?      int32
26  augment /ptp-tt:ptp/ptp-tt:instances/ptp-tt:instance/ptp-tt:ports/ptp-tt:port/
27  ptp-tt:port-ds:
28    +--ro is-measuring-delay?         boolean
29    +--ro as-capable?                 boolean
30    +--rw mean-link-delay-thresh?     ptp-tt:time-interval
31    +--rw neg-mean-link-delay-thresh? ptp-tt:time-interval
32    +--ro neighbor-rate-ratio?        int32
33    +--rw initial-log-announce-interval? int8
34    +--ro current-log-announce-interval? int8
35    +--rw use-mgt-log-announce-interval? boolean
36    +--rw initial-log-sync-interval?   int8
37    +--ro current-log-sync-interval?   int8
38    +--rw use-mgt-log-sync-interval?   boolean
39    +--rw sync-receipt-timeout?        uint8
40    +--ro sync-receipt-timeout-interval? uscaled-ns
41    +--rw initial-log-pdelay-req-interval? int8
42    +--ro current-log-pdelay-req-interval? int8
43    +--rw use-mgt-log-pdelay-req-interval? boolean
44    +--rw mgt-log-pdelay-req-interval?  int8
45    +--rw initial-log-gptp-cap-interval? int8
46    +--ro current-log-gptp-cap-interval? int8
47    +--rw use-mgt-log-gptp-cap-interval? boolean
48    +--rw mgt-log-gptp-cap-interval?    int8
49    +--rw initial-compute-neighbor-rate-ratio? boolean
50    +--ro current-compute-neighbor-rate-ratio? boolean
51    +--rw use-mgt-compute-neighbor-rate-ratio? boolean
52    +--rw mgt-compute-neighbor-rate-ratio? boolean
53    +--rw initial-compute-mean-link-delay? boolean
54    +--ro current-compute-mean-link-delay? boolean
55    +--rw use-mgt-compute-mean-link-delay? boolean

```

```

1  +--rw mgt-compute-mean-link-delay?          boolean
2  +--rw allowed-lost-responses?                uint8
3  +--rw allowed-faults?                      uint8
4  +--rw gptp-cap-receipt-timeout?            uint8
5  +--rw nup?                                float64
6  +--rw ndown?                              float64
7  +--ro one-step-tx-oper?                    boolean
8  +--ro one-step-receive?                   boolean
9  +--ro one-step-transmit?                  boolean
10 +--rw initial-one-step-tx-oper?            boolean
11 +--ro current-one-step-tx-oper?            boolean
12 +--rw use-mgt-one-step-tx-oper?            boolean
13 +--rw mgt-one-step-tx-oper?                boolean
14 +--ro sync-locked?                        boolean
15 +--ro pdelay-truncated-timestamps*         uinteger48
16 +--rw pdelay-req-send-disabled?            boolean
17 +--rw pdelay-resp-send-disabled?            boolean
18 augment /ptp-tt:ptp/ptp-tt:instances/ptp-tt:instance/ptp-tt:ports/ptp-tt:port:
19 +--rw port-statistics-ds
20   +--ro rx-sync-count?                    yang:counter32
21   +--ro rx-one-step-sync-count?            yang:counter32
22   +--ro rx-follow-up-count?                yang:counter32
23   +--ro rx-pdelay-req-count?                yang:counter32
24   +--ro rx-pdelay-resp-count?                yang:counter32
25   +--ro rx-pdelay-resp-follow-up-count?     yang:counter32
26   +--ro rx-announce-count?                 yang:counter32
27   +--ro rx-packet-discard-count?            yang:counter32
28   +--ro sync-receipt-timeout-count?         yang:counter32
29   +--ro announce-receipt-timeout-count?     yang:counter32
30   +--ro pdelay-allowed-lost-exceeded-count? yang:counter32
31   +--ro tx-sync-count?                     yang:counter32
32   +--ro tx-one-step-sync-count?             yang:counter32
33   +--ro tx-follow-up-count?                 yang:counter32
34   +--ro tx-pdelay-req-count?                yang:counter32
35   +--ro tx-pdelay-resp-count?                yang:counter32
36   +--ro tx-pdelay-resp-follow-up-count?     yang:counter32
37   +--ro tx-announce-count?                 yang:counter32
38 augment /ptp-tt:ptp/ptp-tt:instances/ptp-tt:instance/ptp-tt:ports/ptp-tt:port:
39 +--rw asymmetry-measurement-mode-ds
40 +--rw enabled? boolean
41 augment /ptp-tt:ptp/ptp-tt:common-services/ptp-tt:cmls/ptp-tt:ports/ptp-
42 tt:port/ptp-tt:link-port-ds:
43 +--ro cmls-link-port-enabled?              boolean
44 +--ro is-measuring-delay?                  boolean
45 +--ro as-capable-across-domains?           boolean
46 +--rw mean-link-delay-thresh?              ptp-tt:time-interval
47 +--rw neg-mean-link-delay-thresh?          ptp-tt:time-interval
48 +--rw initial-log-pdelay-req-interval?     int8
49 +--ro current-log-pdelay-req-interval?     int8
50 +--rw use-mgt-log-pdelay-req-interval?     boolean
51 +--rw mgt-log-pdelay-req-interval?         int8
52 +--rw initial-compute-neighbor-rate-ratio? boolean
53 +--ro current-compute-neighbor-rate-ratio? boolean
54 +--rw use-mgt-compute-neighbor-rate-ratio? boolean
55 +--rw mgt-compute-neighbor-rate-ratio?     boolean
56 +--rw initial-compute-mean-link-delay?     boolean
57 +--ro current-compute-mean-link-delay?     boolean
58 +--rw use-mgt-compute-mean-link-delay?     boolean
59 +--rw mgt-compute-mean-link-delay?         boolean

```

```

1   +--rw allowed-lost-responses?                uint8
2   +--rw allowed-faults?                        uint8
3   +--ro pdelay-truncated-timestamps*           uinteger48
4   augment /ptp-tt:ptp/ptp-tt:common-services/ptp-tt:cmlDs/ptp-tt:ports/ptp-
5 tt:port:
6   +--rw port-statistics-ds
7       +--ro rx-pdelay-req-count?                yang:counter32
8       +--ro rx-pdelay-resp-count?              yang:counter32
9       +--ro rx-pdelay-resp-follow-up-count?    yang:counter32
10      +--ro rx-packet-discard-count?            yang:counter32
11      +--ro pdelay-allowed-lost-exceeded-count? yang:counter32
12      +--ro tx-pdelay-req-count?                yang:counter32
13      +--ro tx-pdelay-resp-count?              yang:counter32
14      +--ro tx-pdelay-resp-follow-up-count?    yang:counter32
15      augment /ptp-tt:ptp/ptp-tt:common-services/ptp-tt:cmlDs/ptp-tt:ports/ptp-
16 tt:port:
17      +--rw asymmetry-measurement-mode-ds
18      +--rw enabled?    boolean
19

```

20 17.6 YANG modules^{1 2}

21 17.6.1 Module `ieee802-dot1as-gtp`.yang

22 *Replace the YANG module for `ieee802-dot1as-gtp`.yang as follows:*

```

23 module ieee802-dot1as-gtp {
24   yang-version 1.1;
25   namespace "urn:ieee:std:802.1AS:yang:ieee802-dot1as-gtp";
26   prefix dot1as-gtp;
27
28   import ietf-yang-types {
29     prefix yang;
30   }
31   import ieee1588-ptp-tt {
32     prefix ptp-tt;
33   }
34
35   organization
36     "IEEE 802.1 Working Group";
37   contact
38     "WG-URL: http://ieee802.org/1/
39     WG-EMail: stds-802-1-l@ieee.org
40
41     Contact: IEEE 802.1 Working Group Chair
42             Postal: C/O IEEE 802.1 Working Group
43             IEEE Standards Association
44             445 Hoes Lane
45             Piscataway, NJ 08854
46             USA
47
48     E-mail: stds-802-1-chairs@ieee.org";
49   description

```

¹Copyright release for YANG modules: Users of this standard may freely reproduce the YANG modules contained in this subclause so that they can be used for their intended purpose.

²An ASCII version of the YANG modules are attached to the PDF version of this standard, and can be obtained by Web browser from the IEEE 802.1 Website at <https://1.ieee802.org/yang-modules/>.


```

1  "Management objects that control timing and synchronization for time
2  sensitive applications, as specified in clause 14 of IEEE Std
3  802.1AS-2020-REV.
4
5  Copyright (C) IEEE (2025). This version of this YANG module is part
6  of IEEE Std 802.1AS-2020-REV; see the standard itself for full legal
7  notices.";
8
9  revision 2025-03-06 {
10   description
11     "Published as part of IEEE Std 802.1ASds-202X.";
12   reference
13     "IEEE Std 802.1AS - Timing and Synchronization for Time-Sensitive
14     Applications: IEEE Std 802.1AS-2020-REV.
15     IEEE Std 1588 - IEEE Standard for a Precision Clock Synchronization
16     Protocol for Networked Measurement and Control Systems: IEEE Std
17     1588-2019, IEEE Std 1588g-2022, IEEE Std 1588e-2024.";
18 }
19
20 typedef scaled-ns {
21   type string {
22     pattern '[0-9A-F]{2}(-[0-9A-F]{2}){11}';
23   }
24   description
25     "The IEEE Std 802.1AS ScaledNs type represents signed values of
26     time and time interval in units of 2^16 ns, as a signed 96-bit
27     integer. Each of the 12 octets is represented as a pair of
28     hexadecimal characters, using uppercase for a letter. Octets are
29     separated by a dash character. The most significant octet is first.";
30   reference
31     "6.4.3.1 of IEEE Std 802.1AS";
32 }
33
34 typedef unscaled-ns {
35   type string {
36     pattern '[0-9A-F]{2}(-[0-9A-F]{2}){11}';
37   }
38   description
39     "The IEEE Std 802.1AS UScaledNs type represents unsigned values of
40     time and time interval in units of 2^16 ns, as an unsigned 96-bit
41     integer. Each of the 12 octets is represented as a pair of
42     hexadecimal characters, using uppercase for a letter. Octets are
43     separated by a dash character. The most significant octet is first.";
44   reference
45     "6.4.3.2 of IEEE Std 802.1AS";
46 }
47
48 typedef float64 {
49   type string {
50     pattern '[0-9A-F]{2}(-[0-9A-F]{2}){7}';
51   }
52   description
53     "The IEEE Std 802.1AS Float64 type represents IEEE Std 754
54     binary64. Each of the 8 octets is represented as a pair of
55     hexadecimal characters, using uppercase for a letter. Octets are
56     separated by a dash character. The most significant octet is first.";
57   reference
58     "6.4.2 of IEEE Std 802.1AS";
59 }

```

```

1
2 typedef uinteger48 {
3     type uint64 {
4         range "0..281474976710655";
5     }
6     description
7         "48-bit unsigned integer data type.";
8     reference
9         "6.4.2 of IEEE Std 802.1AS";
10 }
11
12 augment "/ptp-tt:ptp/ptp-tt:instances/ptp-tt:instance"
13     + "/ptp-tt:default-ds" {
14     description
15         "Augment IEEE Std 1588 defaultDS.";
16     leaf gm-capable {
17         type boolean;
18         config false;
19         description
20             "The value is true if the time-aware system is capable of being a
21             Grandmaster, and false if the time-aware system is not capable of
22             being a Grandmaster.";
23         reference
24             "14.2.7 of IEEE Std 802.1AS";
25     }
26     leaf current-utc-offset {
27         when "../current-utc-offset-valid='true'";
28         type int16;
29         config false;
30         description
31             "Offset from UTC (TAI - UTC). The offset is in units of seconds.
32             This leaf applies to the ClockTimeTransmitter entity (i.e., local
33             only, unrelated to a remote GM).";
34         reference
35             "14.2.8 of IEEE Std 802.1AS";
36     }
37     leaf current-utc-offset-valid {
38         type boolean;
39         config false;
40         description
41             "The value of current-utc-offset-valid shall be true if the value
42             of current-utc-offset is known to be correct, otherwise it shall
43             be false. This leaf applies to the ClockTimeTransmitter entity
44             (i.e., local only, unrelated to a remote GM).";
45         reference
46             "14.2.9 of IEEE Std 802.1AS";
47     }
48     leaf leap59 {
49         type boolean;
50         config false;
51         description
52             "If the timescale is PTP, a true value for leap59 shall indicate
53             that the last minute of the current UTC day contains 59 seconds.
54             If the timescale is not PTP, the value shall be false. This leaf
55             applies to the ClockTimeTransmitter entity (i.e., local only,
56             unrelated to a remote GM).";
57         reference
58             "14.2.10 of IEEE Std 802.1AS";
59     }

```

```

1  leaf leap61 {
2      type boolean;
3      config false;
4      description
5          "If the timescale is PTP, a true value for leap61 shall indicate
6          that the last minute of the current UTC day contains 61 seconds.
7          If the timescale is not PTP, the value shall be false. This leaf
8          applies to the ClockTimeTransmitter entity (i.e., local only,
9          unrelated to a remote GM).";
10     reference
11         "14.2.11 of IEEE Std 802.1AS";
12 }
13 leaf time-traceable {
14     type boolean;
15     config false;
16     description
17         "The value of time-traceable shall be true if the timescale is
18         traceable to a primary reference; otherwise, the value shall be
19         false. This leaf applies to the ClockTimeTransmitter entity
20         (i.e., local only, unrelated to a remote GM).";
21     reference
22         "14.2.12 of IEEE Std 802.1AS";
23 }
24 leaf frequency-traceable {
25     type boolean;
26     config false;
27     description
28         "The value of frequency-traceable shall be true if the frequency
29         determining the timescale is traceable to a primary reference;
30         otherwise, the value shall be false. This leaf applies to the
31         ClockTimeTransmitter entity (i.e., local only, unrelated to a
32         remote GM).";
33     reference
34         "14.2.13 of IEEE Std 802.1AS";
35 }
36 leaf ptp-timescale {
37     type boolean;
38     config false;
39     description
40         "If ptp-timescale is true, the timescale of the
41         ClockTimeTransmitter entity is PTP, which is the elapsed time
42         since the PTP epoch measured using the second defined by
43         International Atomic Time (TAI). If ptp-timescale is false, the
44         timescale of the ClockTimeTransmitter entity is ARB, which is the
45         elapsed time since an arbitrary epoch. This leaf applies to the
46         ClockTimeTransmitter entity (i.e., local only, unrelated to a
47         remote GM).";
48     reference
49         "14.2.14 of IEEE Std 802.1AS";
50 }
51 leaf time-source {
52     type identityref {
53         base ptp-tt:time-source;
54     }
55     config false;
56     description
57         "The source of time used by the Grandmaster Clock. This leaf
58         applies to the ClockTimeTransmitter entity (i.e., local only,
59         unrelated to a remote GM).";

```

```

1      reference
2      "14.2.15 of IEEE Std 802.1AS";
3  }
4  }
5
6  augment "/ptp-tt:ptp/ptp-tt:instances/ptp-tt:instance"
7      + "/ptp-tt:current-ds" {
8      description
9      "Augment IEEE Std 1588 currentDS.";
10     leaf last-gm-phase-change {
11         type scaled-ns;
12         config false;
13         description
14         "Phase change that occurred on the most recent change in either
15         the Grandmaster PTP Instance or gm-timebase-indicator leaf.";
16         reference
17         "14.3.4 of IEEE Std 802.1AS";
18     }
19     leaf last-gm-freq-change {
20         type float64;
21         config false;
22         description
23         "Frequency change that occurred on the most recent change in
24         either the Grandmaster PTP Instance or gm-timebase-indicator
25         leaf.";
26         reference
27         "14.3.5 of IEEE Std 802.1AS";
28     }
29     leaf gm-timebase-indicator {
30         type uint16;
31         config false;
32         description
33         "The timeBaseIndicator of the current Grandmaster PTP Instance.";
34         reference
35         "14.3.6 of IEEE Std 802.1AS";
36     }
37     leaf gm-change-count {
38         type yang:counter32;
39         config false;
40         description
41         "This statistics counter tracks the number of times the
42         Grandmaster PTP Instance has changed in a gPTP domain.";
43         reference
44         "14.3.7 of IEEE Std 802.1AS";
45     }
46     leaf time-of-last-gm-change {
47         type yang:timestamp;
48         config false;
49         description
50         "System time when the most recent Grandmaster Clock change
51         occurred in a gPTP domain. This leaf's type is YANG timestamp,
52         which is based on system time. System time is an unsigned integer
53         in units of 10 milliseconds, using an epoch defined by the
54         implementation (typically time of boot-up).";
55         reference
56         "14.3.8 of IEEE Std 802.1AS";
57     }
58     leaf time-of-last-phase-change {
59         type yang:timestamp;

```

```

1      config false;
2      description
3          "System time when the most recent change in Grandmaster Clock
4          phase occurred. This leaf's type is YANG timestamp, which is
5          based on system time. System time is an unsigned integer in units
6          of 10 milliseconds, using an epoch defined by the implementation
7          (typically time of boot-up).";
8      reference
9          "14.3.9 of IEEE Std 802.1AS";
10     }
11     leaf time-of-last-freq-change {
12         type yang:timestamp;
13         config false;
14         description
15             "System time when the most recent change in Grandmaster Clock
16             frequency occurred. This leaf's type is YANG timestamp, which is
17             based on system time. System time is an unsigned integer in units
18             of 10 milliseconds, using an epoch defined by the implementation
19             (typically time of boot-up).";
20         reference
21             "14.3.10 of IEEE Std 802.1AS";
22     }
23 }
24
25 augment "/ptp-tt:ptp/ptp-tt:instances/ptp-tt:instance"
26     + "/ptp-tt:parent-ds" {
27     description
28         "Augment IEEE Std 1588 parentDS.";
29     leaf cumulative-rate-ratio {
30         type int32;
31         config false;
32         description
33             "Estimate of the ratio of the frequency of the Grandmaster Clock
34             to the frequency of the LocalClock entity of this PTP Instance.
35             cumulative-rate-ratio is expressed as the fractional frequency
36             offset multiplied by 2^41, i.e., the quantity (rateRatio -
37             1.0)(2^41).";
38         reference
39             "14.4.3 of IEEE Std 802.1AS";
40     }
41 }
42
43 augment "/ptp-tt:ptp"
44     + "/ptp-tt:instances"
45     + "/ptp-tt:instance"
46     + "/ptp-tt:ports"
47     + "/ptp-tt:port"
48     + "/ptp-tt:port-ds" {
49     description
50         "Augment IEEE Std 1588 portDS.
51
52         14.10.4 of IEEE Std 802.1AS specifies ptpPortEnabled
53         (ptp-port-enabled), which is provided in YANG as the semantically
54         equivalent node in ieee1588-ptp-tt named port-enable (in port-ds).
55
56         14.10.15 of IEEE Std 802.1AS specifies
57         mgtSettableLogAnnounceInterval (mgt-log-announce-interval), which
58         is provided in YANG as the semantically equivalent node in
59         ieee1588-ptp-tt named log-announce-interval (in port-ds). In the

```

```

1      context of IEEE Std 802.1AS, log-announce-interval cannot be used
2      unless use-mgt-log-announce-interval is true.
3
4      14.10.20 of IEEE Std 802.1AS specifies mgtSettableLogSyncInterval
5      (mgt-log-sync-interval), which is provided in YANG as the
6      semantically equivalent node in ieee1588-ptp-tt named
7      log-sync-interval (in port-ds). In the context of IEEE Std 802.1AS,
8      log-sync-interval cannot be used unless use-mgt-log-sync-interval
9      is true."
10     leaf is-measuring-delay {
11         type boolean;
12         config false;
13         description
14             "Boolean that is true if the port is measuring PTP Link
15             propagation delay.";
16         reference
17             "14.10.6 of IEEE Std 802.1AS";
18     }
19     leaf as-capable {
20         type boolean;
21         config false;
22         description
23             "Boolean that is true if and only if it is determined that this
24             PTP Instance and the PTP Instance at the other end of the link
25             attached to this port can interoperate with each other via the
26             IEEE Std 802.1AS protocol.";
27         reference
28             "10.2.5.1 of IEEE Std 802.1AS
29             14.10.7 of IEEE Std 802.1AS";
30     }
31     leaf mean-link-delay-thresh {
32         type ptp-tt:time-interval;
33         description
34             "Propagation time threshold for mean-link-delay, above which a
35             port is not considered capable of participating in the IEEE Std
36             802.1AS protocol.";
37         reference
38             "14.10.9 of IEEE Std 802.1AS";
39     }
40     leaf neg-mean-link-delay-thresh {
41         type ptp-tt:time-interval;
42         description
43             "The negative propagation time threshold for mean-link-delay,
44             below which a port is not considered capable of participating
45             in the IEEE Std 802.1AS protocol.";
46         reference
47             "14.10.10 of IEEE Std 802.1AS";
48     }
49     leaf neighbor-rate-ratio {
50         type int32;
51         config false;
52         description
53             "Estimate of the ratio of the frequency of the LocalClock entity
54             of the PTP Instance at the other end of the link attached to this
55             PTP Port, to the frequency of the LocalClock entity of this PTP
56             Instance. neighbor-rate-ratio is expressed as the fractional
57             frequency offset multiplied by 2^41, i.e., the quantity
58             (rateRatio - 1.0)(2^41).";
59         reference

```

```

1      "14.10.12 of IEEE Std 802.1AS";
2  }
3  leaf initial-log-announce-interval {
4      type int8;
5      description
6          "When use-mgt-log-announce-interval is false (i.e., change with
7           Signaling message), this is the logarithm to base 2 of the
8           announce interval used when the port is initialized.";
9      reference
10         "14.10.13 of IEEE Std 802.1AS";
11  }
12  leaf current-log-announce-interval {
13      type int8;
14      config false;
15      description
16          "Logarithm to base 2 of the current announce interval.";
17      reference
18          "14.10.14 of IEEE Std 802.1AS";
19  }
20  leaf use-mgt-log-announce-interval {
21      type boolean;
22      description
23          "Boolean that determines the source of the announce interval. If
24           the value is true, the announce interval
25           (current-log-announce-interval) is set equal to the value of
26           mgt-log-announce-interval. If the value is false, the announce
27           interval is determined by the AnnounceIntervalSetting state
28           machine (i.e., changed with Signaling message).";
29      reference
30          "14.10.15 of IEEE Std 802.1AS";
31  }
32  leaf initial-log-sync-interval {
33      type int8;
34      description
35          "When use-mgt-log-sync-interval is false (i.e., change with
36           Signaling message), this is the logarithm to base 2 of the
37           sync interval used when the port is initialized.";
38      reference
39          "14.10.17 of IEEE Std 802.1AS";
40  }
41  leaf current-log-sync-interval {
42      type int8;
43      config false;
44      description
45          "Logarithm to base 2 of the current sync interval.";
46      reference
47          "14.10.18 of IEEE Std 802.1AS";
48  }
49  leaf use-mgt-log-sync-interval {
50      type boolean;
51      description
52          "Boolean that determines the source of the sync interval. If the
53           value is true, the sync interval (current-log-sync-interval) is
54           set equal to the value of mgt-log-sync-interval. If the value is
55           false, the sync interval is determined by the SyncIntervalSetting
56           state machine (i.e., changed with Signaling message).";
57      reference
58          "14.10.19 of IEEE Std 802.1AS";
59  }

```

```

1  leaf sync-receipt-timeout {
2      type uint8;
3      description
4          "Number of sync intervals that a timeReceiver port waits without
5          receiving synchronization information, before assuming that the
6          timeTransmitter is no longer transmitting synchronization
7          information and that the BTCA needs to be run, if appropriate.";
8      reference
9          "14.10.21 of IEEE Std 802.1AS";
10 }
11 leaf sync-receipt-timeout-interval {
12     type uscaled-ns;
13     config false;
14     description
15         "Time interval after which sync receipt timeout occurs if
16         time-synchronization information has not been received during the
17         interval.";
18     reference
19         "14.10.22 of IEEE Std 802.1AS";
20 }
21 leaf initial-log-pdelay-req-interval {
22     type int8;
23     description
24         "When use-mgt-log-pdelay-req-interval is false (i.e., change with
25         Signaling message), this is the logarithm to base 2 of the
26         Pdelay_Req transmit interval used when the port is initialized.";
27     reference
28         "14.10.23 of IEEE Std 802.1AS";
29 }
30 leaf current-log-pdelay-req-interval {
31     type int8;
32     config false;
33     description
34         "Logarithm to base 2 of the current Pdelay_Req transmit interval.";
35     reference
36         "14.10.24 of IEEE Std 802.1AS";
37 }
38 leaf use-mgt-log-pdelay-req-interval {
39     type boolean;
40     description
41         "Boolean that determines the source of the Pdelay_Req transmit
42         interval. If the value is true, the Pdelay_Req transmit interval
43         (current-log-pdelay-req-interval) is set equal to the value of
44         mgt-log-pdelay-req-interval. If the value is false, the
45         Pdelay_Req transmit interval is determined by the
46         LinkDelayIntervalSetting state machine (i.e., changed with
47         Signaling message).";
48     reference
49         "14.10.25 of IEEE Std 802.1AS";
50 }
51 leaf mgt-log-pdelay-req-interval {
52     type int8;
53     description
54         "Logarithm to base 2 of the Pdelay_Req transmit interval, used if
55         use-mgt-log-pdelay-req-interval is true. This value is not used
56         if use-mgt-log-pdelay-req-interval is false.";
57     reference
58         "14.10.26 of IEEE Std 802.1AS";
59 }

```



```

1  leaf initial-log-gtp-cap-interval {
2      type int8;
3      description
4          "When use-mgt-log-gtp-cap-interval is false (i.e., change with
5          Signaling message), this is the logarithm to base 2 of the
6          gTP capable message interval used when the port is initialized.";
7      reference
8          "14.10.27 of IEEE Std 802.1AS";
9  }
10 leaf current-log-gtp-cap-interval {
11     type int8;
12     config false;
13     description
14         "Logarithm to base 2 of the current gTP capable message
15         interval.";
16     reference
17         "14.10.28 of IEEE Std 802.1AS";
18 }
19 leaf use-mgt-log-gtp-cap-interval {
20     type boolean;
21     description
22         "Boolean that determines the source of the gTP capable message
23         interval. If the value is true, the gTP capable message interval
24         (current-log-gtp-cap-interval) is set equal to the value of
25         mgt-gtp-cap-req-interval. If the value is false, the gTP
26         capable message interval is determined by the
27         GtpCapableMessageIntervalSetting state machine (i.e., changed
28         with Signaling message).";
29     reference
30         "14.10.29 of IEEE Std 802.1AS";
31 }
32 leaf mgt-log-gtp-cap-interval {
33     type int8;
34     description
35         "Logarithm to base 2 of the gTP capable message interval, used
36         if use-mgt-log-gtp-cap-interval is true. This value is not used
37         if use-mgt-log-pdelay-req-interval is false.";
38     reference
39         "14.10.30 of IEEE Std 802.1AS";
40 }
41 leaf initial-compute-neighbor-rate-ratio {
42     type boolean;
43     description
44         "When use-mgt-compute-neighbor-rate-ratio is false (i.e., change
45         with Signaling message), this is the initial value of
46         computeNeighborRateRatio.";
47     reference
48         "14.10.31 of IEEE Std 802.1AS";
49 }
50 leaf current-compute-neighbor-rate-ratio {
51     type boolean;
52     config false;
53     description
54         "Current value of computeNeighborRateRatio.";
55     reference
56         "14.10.32 of IEEE Std 802.1AS";
57 }
58 leaf use-mgt-compute-neighbor-rate-ratio {
59     type boolean;

```

```

1      description
2      "Boolean that determines the source of computeNeighborRateRatio..
3      If the value is true, computeNeighborRateRatio is set equal to
4      the value of mgt-compute-neighbor-rate-ratio. If the value is
5      false, computeNeighborRateRatio is determined by the
6      LinkDelayIntervalSetting state machine (i.e., changed with
7      Signaling message).";
8      reference
9      "14.10.33 of IEEE Std 802.1AS";
10     }
11     leaf mgt-compute-neighbor-rate-ratio {
12         type boolean;
13         description
14         "Value of computeNeighborRateRatio, used if
15         use-mgt-compute-neighbor-rate-ratio is true. This value is not
16         used if use-mgt-compute-neighbor-rate-ratio is false.";
17         reference
18         "14.10.34 of IEEE Std 802.1AS";
19     }
20     leaf initial-compute-mean-link-delay {
21         type boolean;
22         description
23         "When use-mgt-compute-mean-link-delay is false (i.e., change with
24         Signaling message), this is the initial value of
25         computeMeanLinkDelay.";
26         reference
27         "14.10.35 of IEEE Std 802.1AS";
28     }
29     leaf current-compute-mean-link-delay {
30         type boolean;
31         config false;
32         description
33         "Current value of computeMeanLinkDelay.";
34         reference
35         "14.10.36 of IEEE Std 802.1AS";
36     }
37     leaf use-mgt-compute-mean-link-delay {
38         type boolean;
39         description
40         "Boolean that determines the source of computeMeanLinkDelay. If
41         the value is true, computeMeanLinkDelay is set equal to the value
42         of mgt-compute-mean-link-delay. If the value is false,
43         computeMeanLinkDelay is determined by the
44         LinkDelayIntervalSetting state machine (i.e., changed with
45         Signaling message).";
46         reference
47         "14.10.37 of IEEE Std 802.1AS";
48     }
49     leaf mgt-compute-mean-link-delay {
50         type boolean;
51         description
52         "Value of computeMeanLinkDelay, used if
53         use-mgt-compute-mean-link-delay is true. This value is not used
54         if use-mgt-compute-mean-link-delay is false.";
55         reference
56         "14.10.38 of IEEE Std 802.1AS";
57     }
58     leaf allowed-lost-responses {
59         type uint8;

```

```

1      description
2          "Number of Pdelay_Req messages for which a valid response is not
3          received, above which a port is considered to not be exchanging
4          peer delay messages with its neighbor.";
5      reference
6          "14.10.39 of IEEE Std 802.1AS";
7  }
8  leaf allowed-faults {
9      type uint8;
10     description
11         "Number of faults above which asCapable is set to false.";
12     reference
13         "14.10.40 of IEEE Std 802.1AS";
14 }
15 leaf gptp-cap-receipt-timeout {
16     type uint8;
17     description
18         "Number of transmission intervals that a port waits without
19         receiving the gPTP capable TLV, before assuming that the neighbor
20         port is no longer invoking the gPTP protocol.";
21     reference
22         "14.10.41 of IEEE Std 802.1AS";
23 }
24 leaf nup {
25     type float64;
26     description
27         "For an OLT port of an IEEE Std 802.3 EPON link, this value is
28         the effective index of refraction for the EPON upstream
29         wavelength light of the optical path.";
30     reference
31         "14.10.43 of IEEE Std 802.1AS";
32 }
33 leaf ndown {
34     type float64;
35     description
36         "For an OLT port of an IEEE 802.3 EPON link, this value is the
37         effective index of refraction for the EPON downstream wavelength
38         light of the optical path.";
39     reference
40         "14.10.44 of IEEE Std 802.1AS";
41 }
42 leaf one-step-tx-oper {
43     type boolean;
44     config false;
45     description
46         "This value is true if the port is sending one-step Sync
47         messages, and false if the port is sending two-step Sync and
48         Follow-Up messages.";
49     reference
50         "14.10.45 of IEEE Std 802.1AS";
51 }
52 leaf one-step-receive {
53     type boolean;
54     config false;
55     description
56         "This value is true if the port is capable of receiving and
57         processing one-step Sync messages.";
58     reference
59         "14.10.46 of IEEE Std 802.1AS";

```

```

1   }
2   leaf one-step-transmit {
3       type boolean;
4       config false;
5       description
6           "This value is true if the port is capable of transmitting
7           one-step Sync messages.";
8       reference
9           "14.10.47 of IEEE Std 802.1AS";
10  }
11  leaf initial-one-step-tx-oper {
12      type boolean;
13      description
14          "When use-mgt-one-step-tx-oper is false (i.e., change with
15          Signaling message), this is the initial value of
16          current-one-step-tx-oper.";
17      reference
18          "14.10.48 of IEEE Std 802.1AS";
19  }
20  leaf current-one-step-tx-oper {
21      type boolean;
22      config false;
23      description
24          "This value is true if the port is configured to transmit
25          one-step Sync messages, either via management
26          (mgt-one-step-tx-oper) or Signaling. If both
27          current-one-step-tx-oper and one-step-transmit are true, the port
28          transmits one-step Sync messages (i.e., one-step-tx-oper true).";
29      reference
30          "14.10.49 of IEEE Std 802.1AS";
31  }
32  leaf use-mgt-one-step-tx-oper {
33      type boolean;
34      description
35          "Boolean that determines the source of current-one-step-tx-oper.
36          If the value is true, current-one-step-tx-oper is set equal to
37          the value of mgt-one-step-tx-oper. If the value is false,
38          current-one-step-tx-oper is determined by the
39          OneStepTxOperSetting state machine (i.e., changed with Signaling
40          message).";
41      reference
42          "14.10.50 of IEEE Std 802.1AS";
43  }
44  leaf mgt-one-step-tx-oper {
45      type boolean;
46      description
47          "If use-mgt-one-step-tx-oper is true, current-one-step-tx-oper is
48          set equal to this value. This value is not used if
49          use-mgt-one-step-tx-oper is false.";
50      reference
51          "14.10.51 of IEEE Std 802.1AS";
52  }
53  leaf sync-locked {
54      type boolean;
55      config false;
56      description
57          "This value is true if the port will transmit a Sync as soon as
58          possible after the timeReceiver port receives a Sync message.";
59      reference

```

```

1      "14.10.52 of IEEE Std 802.1AS";
2  }
3  leaf-list pdelay-truncated-timestamps {
4      type uinteger48;
5      config false;
6      description
7          "For full-duplex IEEE Std 802.3 media, and CSN media that use the
8          peer-to-peer delay mechanism to measure path delay, the values of
9          the four elements of this leaf-list correspond to the timestamps
10         t1, t2, t3, and t4, listed in that order. Each timestamp is
11         expressed in units of 2-16 ns (i.e., the value of each array
12         element is equal to the remainder obtained upon dividing the
13         respective timestamp, expressed in units of 2-16 ns, by 248).
14         At any given time, the timestamp values stored in the array are
15         for the same, and most recently completed, peer delay message
16         exchange. For each timestamp, only 48-bits are valid (the upper
17         16-bits are always zero).";
18     reference
19         "14.10.53 of IEEE Std 802.1AS";
20 }
21 leaf pdelay-req-send-disabled {
22     type boolean;
23     description
24         "A boolean that is administratively set to TRUE
25         if Pdelay_Req messages are not transmitted by this port.
26         The default value for this variable shall be FALSE.";
27     reference
28         "11.2.19.2.14 of IEEE Std 802.1ASds";
29 }
30 leaf pdelay-resp-send-disabled {
31     type boolean;
32     description
33         "A boolean that is administratively set to TRUE
34         if Pdelay_Resp messages are not transmitted by this port.
35         The default value for this variable shall be FALSE.";
36     reference
37         "11.2.20.2.6 of IEEE Std 802.1ASds";
38 }
39 }
40
41 augment "/ptp-tt:ptp"
42     + "/ptp-tt:instances"
43     + "/ptp-tt:instance"
44     + "/ptp-tt:ports"
45     + "/ptp-tt:port" {
46     description
47         "Augment to add port-statistics-ds to IEEE Std 1588 PTP Port.";
48     container port-statistics-ds {
49         description
50             "Provides counters associated with the port of the PTP Instance.";
51         reference
52             "14.12 of IEEE Std 802.1AS";
53         leaf rx-sync-count {
54             type yang:counter32;
55             config false;
56             description
57                 "Counter that increments every time synchronization information
58                 is received.";
59             reference

```

```

1      "14.12.2 of IEEE Std 802.1AS";
2  }
3  leaf rx-one-step-sync-count {
4      type yang:counter32;
5      config false;
6      description
7          "Counter that increments every time a one-step Sync message is
8          received.";
9      reference
10         "14.12.3 of IEEE Std 802.1AS";
11 }
12 leaf rx-follow-up-count {
13     type yang:counter32;
14     config false;
15     description
16         "Counter that increments every time a Follow_Up message is
17         received.";
18     reference
19         "14.12.4 of IEEE Std 802.1AS";
20 }
21 leaf rx-pdelay-req-count {
22     type yang:counter32;
23     config false;
24     description
25         "Counter that increments every time a Pdelay_Req message is
26         received.";
27     reference
28         "14.12.5 of IEEE Std 802.1AS";
29 }
30 leaf rx-pdelay-resp-count {
31     type yang:counter32;
32     config false;
33     description
34         "Counter that increments every time a Pdelay_Resp message is
35         received.";
36     reference
37         "14.12.6 of IEEE Std 802.1AS";
38 }
39 leaf rx-pdelay-resp-follow-up-count {
40     type yang:counter32;
41     config false;
42     description
43         "Counter that increments every time a Pdelay_Resp_Follow_Up
44         message is received.";
45     reference
46         "14.12.7 of IEEE Std 802.1AS";
47 }
48 leaf rx-announce-count {
49     type yang:counter32;
50     config false;
51     description
52         "Counter that increments every time an Announce message is
53         received.";
54     reference
55         "14.12.8 of IEEE Std 802.1AS";
56 }
57 leaf rx-packet-discard-count {
58     type yang:counter32;
59     config false;

```

```

1      description
2      "Counter that increments every time a PTP message of the
3      respective PTP Instance is discarded.";
4      reference
5      "14.12.9 of IEEE Std 802.1AS";
6  }
7  leaf sync-receipt-timeout-count {
8      type yang:counter32;
9      config false;
10     description
11     "Counter that increments every time a sync receipt timeout
12     occurs.";
13     reference
14     "14.12.10 of IEEE Std 802.1AS";
15 }
16 leaf announce-receipt-timeout-count {
17     type yang:counter32;
18     config false;
19     description
20     "Counter that increments every time an announce receipt timeout
21     occurs.";
22     reference
23     "14.12.11 of IEEE Std 802.1AS";
24 }
25 leaf pdelay-allowed-lost-exceeded-count {
26     type yang:counter32;
27     config false;
28     description
29     "Counter that increments every time the value of the variable
30     lostResponses exceeds the value of the variable
31     allowedLostResponses, in the RESET state of the MDPdelayReq
32     state machine.";
33     reference
34     "14.12.12 of IEEE Std 802.1AS";
35 }
36 leaf tx-sync-count {
37     type yang:counter32;
38     config false;
39     description
40     "Counter that increments every time synchronization information
41     is transmitted.";
42     reference
43     "14.12.13 of IEEE Std 802.1AS";
44 }
45 leaf tx-one-step-sync-count {
46     type yang:counter32;
47     config false;
48     description
49     "Counter that increments every time a one-step Sync message is
50     transmitted.";
51     reference
52     "14.12.14 of IEEE Std 802.1AS";
53 }
54 leaf tx-follow-up-count {
55     type yang:counter32;
56     config false;
57     description
58     "Counter that increments every time a Follow_Up message is
59     transmitted.";

```

```

1      reference
2      "14.12.15 of IEEE Std 802.1AS";
3  }
4  leaf tx-pdelay-req-count {
5      type yang:counter32;
6      config false;
7      description
8          "Counter that increments every time a Pdelay_Req message is
9          transmitted.";
10     reference
11         "14.12.16 of IEEE Std 802.1AS";
12 }
13 leaf tx-pdelay-resp-count {
14     type yang:counter32;
15     config false;
16     description
17         "Counter that increments every time a Pdelay_Resp message is
18         transmitted.";
19     reference
20         "14.12.17 of IEEE Std 802.1AS";
21 }
22 leaf tx-pdelay-resp-follow-up-count {
23     type yang:counter32;
24     config false;
25     description
26         "Counter that increments every time a Pdelay_Resp_Follow_Up
27         message is transmitted.";
28     reference
29         "14.12.18 of IEEE Std 802.1AS";
30 }
31 leaf tx-announce-count {
32     type yang:counter32;
33     config false;
34     description
35         "Counter that increments every time an Announce message is
36         transmitted.";
37     reference
38         "14.12.19 of IEEE Std 802.1AS";
39 }
40 }
41 }
42
43 augment "/ptp-tt:ptp"
44     + "/ptp-tt:instances"
45     + "/ptp-tt:instance"
46     + "/ptp-tt:ports"
47     + "/ptp-tt:port" {
48     description
49         "Augment to add asymmetry-measurement-mode-ds to IEEE Std 1588 PTP
50         Port.";
51     container asymmetry-measurement-mode-ds {
52         description
53             "Represents the capability to enable/disable the Asymmetry
54             Compensation Measurement Procedure on a PTP Port. This data set
55             is used instead of the CMLDS asymmetry-measurement-mode-ds when
56             only a single PTP Instance is present (i.e., CMLDS is not used).";
57         reference
58             "14.15 of IEEE Std 802.1AS
59             Annex G of IEEE Std 802.1AS";

```



```

1     leaf enabled {
2         type boolean;
3         description
4             "For full-duplex IEEE Std 802.3 media, the value is true if an
5             asymmetry measurement is being performed for the link attached
6             to this PTP Port, and false otherwise. For all other media, the
7             value shall be false.";
8         reference
9             "14.15.2 of IEEE Std 802.1AS";
10    }
11 }
12 }
13
14 augment "/ptp-tt:ptp"
15     + "/ptp-tt:common-services"
16     + "/ptp-tt:cmlds"
17     + "/ptp-tt:ports"
18     + "/ptp-tt:port"
19     + "/ptp-tt:link-port-ds" {
20     description
21         "Augment IEEE Std 1588 cmldsLinkPortDS.";
22     leaf cmlds-link-port-enabled {
23         type boolean;
24         config false;
25         description
26             "Boolean that is true if both delay-mechanism is common-p2p and
27             the value of ptp-port-enabled is true, for at least one PTP Port
28             that uses the CMLDS; otherwise, the value is false.";
29         reference
30             "11.2.18.1 of IEEE Std 802.1AS
31             14.18.3 of IEEE Std 802.1AS";
32     }
33     leaf is-measuring-delay {
34         type boolean;
35         config false;
36         description
37             "This leaf is analogous to is-measuring-delay for a PTP Port, but
38             applicable to this Link Port.";
39         reference
40             "14.18.4 of IEEE Std 802.1AS";
41     }
42     leaf as-capable-across-domains {
43         type boolean;
44         config false;
45         description
46             "This leaf is true when all PTP Instances (domains) for this Link
47             Port detect proper exchange of Pdelay messages.";
48         reference
49             "11.2.2 of IEEE Std 802.1AS
50             14.18.5 of IEEE Std 802.1AS";
51     }
52     leaf mean-link-delay-thresh {
53         type ptp-tt:time-interval;
54         description
55             "Propagation time threshold for mean-link-delay, above which a
56             Link Port is not considered capable of participating in the IEEE
57             Std 802.1AS protocol.";
58         reference
59             "14.18.7 of IEEE Std 802.1AS";

```

```

1      }
2      leaf neg-mean-link-delay-thresh {
3          type ptp-tt:time-interval;
4          description
5              "The negative propagation time threshold for mean-link-delay,
6               below which a Link Port is not considered capable of
7               participating in the IEEE Std 802.1AS protocol.";
8          reference
9              "14.18.8 of IEEE Std 802.1AS";
10     }
11     leaf initial-log-pdelay-req-interval {
12         type int8;
13         description
14             "This leaf is analogous to initial-log-pdelay-req-interval for a
15              PTP Port, but applicable to this Link Port.";
16         reference
17             "14.18.11 of IEEE Std 802.1AS";
18     }
19     leaf current-log-pdelay-req-interval {
20         type int8;
21         config false;
22         description
23             "This leaf is analogous to current-log-pdelay-req-interval for a
24              PTP Port, but applicable to this Link Port.";
25         reference
26             "14.18.12 of IEEE Std 802.1AS";
27     }
28     leaf use-mgt-log-pdelay-req-interval {
29         type boolean;
30         description
31             "This leaf is analogous to use-mgt-log-pdelay-req-interval for a
32              PTP Port, but applicable to this Link Port.";
33         reference
34             "14.18.13 of IEEE Std 802.1AS";
35     }
36     leaf mgt-log-pdelay-req-interval {
37         type int8;
38         description
39             "This leaf is analogous to mgt-log-pdelay-req-interval for a PTP
40              Port, but applicable to this Link Port.";
41         reference
42             "14.18.14 of IEEE Std 802.1AS";
43     }
44     leaf initial-compute-neighbor-rate-ratio {
45         type boolean;
46         description
47             "This leaf is analogous to initial-compute-neighbor-rate-ratio
48              for a PTP Port, but applicable to this Link Port.";
49         reference
50             "14.18.15 of IEEE Std 802.1AS";
51     }
52     leaf current-compute-neighbor-rate-ratio {
53         type boolean;
54         config false;
55         description
56             "This leaf is analogous to current-compute-neighbor-rate-ratio
57              for a PTP Port, but applicable to this Link Port.";
58         reference
59             "14.18.16 of IEEE Std 802.1AS";

```

```

1   }
2   leaf use-mgt-compute-neighbor-rate-ratio {
3       type boolean;
4       description
5           "This leaf is analogous to use-mgt-compute-neighbor-rate-ratio
6           for a PTP Port, but applicable to this Link Port.";
7       reference
8           "14.18.17 of IEEE Std 802.1AS";
9   }
10  leaf mgt-compute-neighbor-rate-ratio {
11      type boolean;
12      description
13          "This leaf is analogous to mgt-compute-neighbor-rate-ratio for a
14          PTP Port, but applicable to this Link Port.";
15      reference
16          "14.18.18 of IEEE Std 802.1AS";
17  }
18  leaf initial-compute-mean-link-delay {
19      type boolean;
20      description
21          "This leaf is analogous to initial-compute-mean-link-delay for a
22          PTP Port, but applicable to this Link Port.";
23      reference
24          "14.18.19 of IEEE Std 802.1AS";
25  }
26  leaf current-compute-mean-link-delay {
27      type boolean;
28      config false;
29      description
30          "This leaf is analogous to current-compute-mean-link-delay for a
31          PTP Port, but applicable to this Link Port.";
32      reference
33          "14.18.20 of IEEE Std 802.1AS";
34  }
35  leaf use-mgt-compute-mean-link-delay {
36      type boolean;
37      description
38          "This leaf is analogous to use-mgt-compute-mean-link-delay for a
39          PTP Port, but applicable to this Link Port.";
40      reference
41          "14.18.21 of IEEE Std 802.1AS";
42  }
43  leaf mgt-compute-mean-link-delay {
44      type boolean;
45      description
46          "This leaf is analogous to mgt-compute-mean-link-delay for a PTP
47          Port, but applicable to this Link Port.";
48      reference
49          "14.18.22 of IEEE Std 802.1AS";
50  }
51  leaf allowed-lost-responses {
52      type uint8;
53      description
54          "This leaf is analogous to allowed-lost-responses for a PTP Port,
55          but applicable to this Link Port.";
56      reference
57          "14.18.23 of IEEE Std 802.1AS";
58  }
59  leaf allowed-faults {

```

```

1      type uint8;
2      description
3          "This leaf is analogous to allowed-faults for a PTP Port, but
4          applicable to this Link Port.";
5      reference
6          "14.18.24 of IEEE Std 802.1AS";
7  }
8  leaf-list pdelay-truncated-timestamps {
9      type uinteger48;
10     config false;
11     description
12         "This leaf is analogous to pdelay-truncated-timestamps for a PTP
13         Port, but applicable to this Link Port.";
14     reference
15         "14.18.26 of IEEE Std 802.1AS";
16 }
17 }
18
19 augment "/ptp-tt:ptp"
20     + "/ptp-tt:common-services"
21     + "/ptp-tt:cmllds"
22     + "/ptp-tt:ports"
23     + "/ptp-tt:port" {
24     description
25         "Augment to add port-statistics-ds to IEEE Std 1588 Link Port.";
26     container port-statistics-ds {
27         description
28             "This container is analogous to port-statistics-ds for a PTP
29             Port, but applicable to this Link Port.";
30         reference
31             "14.19 of IEEE Std 802.1AS";
32         leaf rx-pdelay-req-count {
33             type yang:counter32;
34             config false;
35             description
36                 "This leaf is analogous to rx-pdelay-req-count for a PTP Port,
37                 but applicable to this Link Port.";
38             reference
39                 "14.19.2 of IEEE Std 802.1AS";
40         }
41         leaf rx-pdelay-resp-count {
42             type yang:counter32;
43             config false;
44             description
45                 "This leaf is analogous to rx-pdelay-resp-count for a PTP Port,
46                 but applicable to this Link Port.";
47             reference
48                 "14.19.3 of IEEE Std 802.1AS";
49         }
50         leaf rx-pdelay-resp-follow-up-count {
51             type yang:counter32;
52             config false;
53             description
54                 "This leaf is analogous to rx-pdelay-resp-follow-up-count for a
55                 PTP Port, but applicable to this Link Port.";
56             reference
57                 "14.19.4 of IEEE Std 802.1AS";
58         }
59         leaf rx-packet-discard-count {

```

```

1      type yang:counter32;
2      config false;
3      description
4          "This leaf is analogous to rx-packet-discard-count for a PTP
5          Port, but applicable to this Link Port.";
6      reference
7          "14.19.5 of IEEE Std 802.1AS";
8  }
9  leaf pdelay-allowed-lost-exceeded-count {
10     type yang:counter32;
11     config false;
12     description
13         "This leaf is analogous to pdelay-allowed-lost-exceeded-count
14         for a PTP Port, but applicable to this Link Port.";
15     reference
16         "14.19.6 of IEEE Std 802.1AS";
17 }
18 leaf tx-pdelay-req-count {
19     type yang:counter32;
20     config false;
21     description
22         "This leaf is analogous to tx-pdelay-req-count for a PTP Port,
23         but applicable to this Link Port.";
24     reference
25         "14.19.7 of IEEE Std 802.1AS";
26 }
27 leaf tx-pdelay-resp-count {
28     type yang:counter32;
29     config false;
30     description
31         "This leaf is analogous to tx-pdelay-resp-count for a PTP Port,
32         but applicable to this Link Port.";
33     reference
34         "14.19.8 of IEEE Std 802.1AS";
35 }
36 leaf tx-pdelay-resp-follow-up-count {
37     type yang:counter32;
38     config false;
39     description
40         "This leaf is analogous to tx-pdelay-resp-follow-up-count for a
41         PTP Port, but applicable to this Link Port.";
42     reference
43         "14.19.10 of IEEE Std 802.1AS";
44 }
45 }
46 }
47
48 augment "/ptp-tt:ptp"
49     + "/ptp-tt:common-services"
50     + "/ptp-tt:cmls"
51     + "/ptp-tt:ports"
52     + "/ptp-tt:port" {
53     description
54         "Augment to add asymmetry-measurement-mode-ds to IEEE Std 1588 Link
55         Port.";
56     container asymmetry-measurement-mode-ds {
57         description
58             "This container is analogous to asymmetry-measurement-mode-ds for
59             a PTP Port, but applicable to this Link Port.";

```

```
1      reference
2      "14.20 of IEEE Std 802.1AS";
3      leaf enabled {
4          type boolean;
5          description
6              "This leaf is analogous to
7              asymmetry-measurement-mode-ds.enabled for a PTP Port, but
8              applicable to this Link Port.";
9          reference
10             "14.20.2 of IEEE Std 802.1AS";
11     }
12 }
13 }
14 }
15
16
```

1 *Insert the following new Clause 19:*

2 **19. Media-dependent layer specification for IEEE Std 802.3 Clause 4 Media**
3 **Access Control (MAC) operating in half-duplex**

4 **19.1 Overview**

5 **19.1.1 General**

6 This clause specifies the media-dependent layer that provides synchronized time across links using the IEEE
7 Std 802.3 Clause 4 MAC operating in half-duplex mode

8 **19.1.1.1 Half-duplex Ethernet (HDE) characteristics**

9 The IEEE Std 802.3 Clause 4 MAC can operate in either full-duplex or half-duplex mode. When this MAC
10 is operating in full-duplex, its media-dependent specification for gPTP is covered in Clause 11. Clause 19 is
11 used when the IEEE Std 802.3 Clause 4 MAC is operating in half-duplex. This mode necessitates additional
12 managed object settings and frame processing due to the effects of the shared media this mode supports,
13 which are described in this clause.

14 Figure 19-1 shows an example of time-aware network using HDE links.

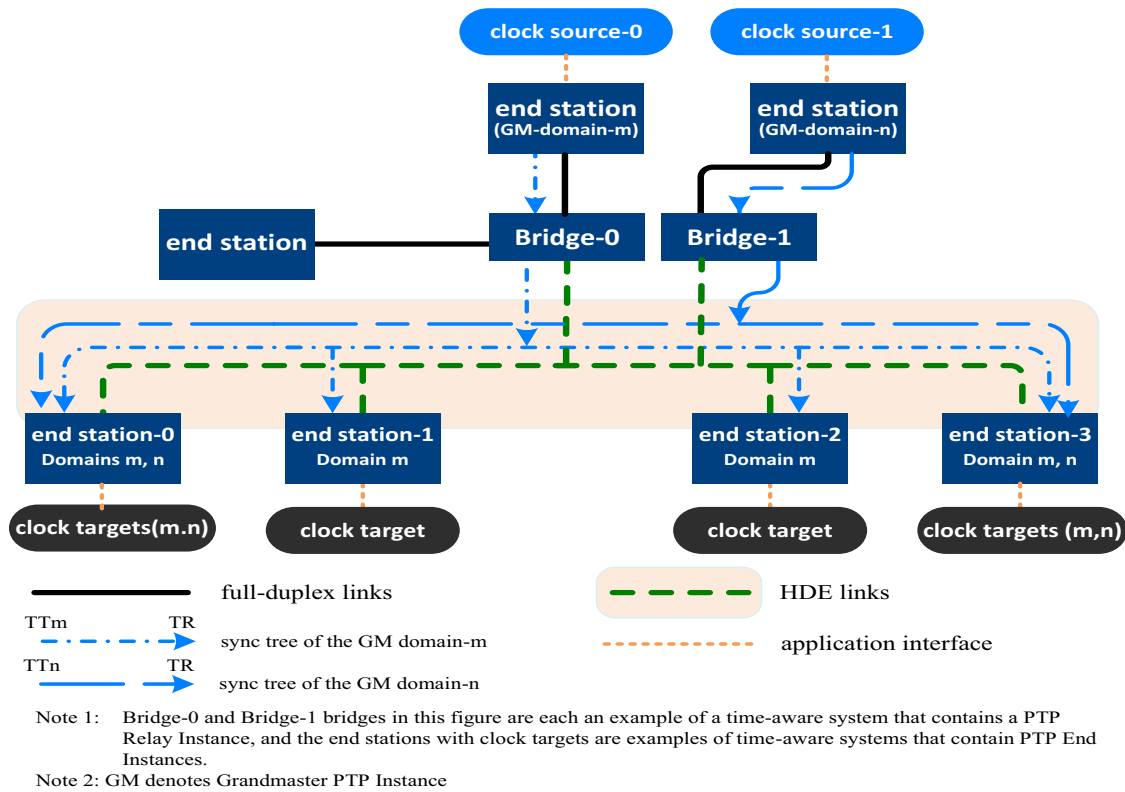


Figure 19-1— Time-aware network example where the four end stations near the bottom of the figure are connected to the bridge via an HDE link

1 When using HDE links, the use of “neighbor” refers to a device or to devices (end station or bridge) that is/

2 are the intended recipient(s) of a transmitted PTP message. The intended recipient(s) of a PTP message from

3 a gPTP TT port is/are one or more gPTP TR port(s). The intended recipient of a PTP message from a gPTP

4 TR port is the gPTP TT port, regardless of whether or not other gPTP TT ports received that message (e.g.,

5 if all ports are connected via a shared medium). In the example in Figure 19-1, PTP messages are exchanged

6 between a gPTP TT port of a bridge (e.g., Bridge-0) and the four gPTP TR ports of the four end stations via

7 a shared medium.

8 NOTE—For example, this standard supports 10BASE-T1S, as specified in Clause 147 of IEEE Std 802.3, utilizing

9 Physical Layer Collision Avoidance (PLCA), as specified in Clause 148 of IEEE Std 802.3.

10 **19.1.1.2 Overview of the major differences and restrictions of using HDE links**

11 This standard uses Clause 11 in its entirety with the following differences (complete details follow starting

12 in 19.1.2):

- 13 a) The peer delay initiator is restricted to timeReceivers only (see 19.1.2).
- 14 b) External port configuration mode is the only mode supported (see 19.8).

15 The following features of this standard have not been qualified for use with HDE links:

- 16 c) One-step time transport (see 19.1.3 and 19.2.16)
- 17 d) CMLDS (see 19.2.17)
- 18 e) The use of Signaling messages (see 19.8)
- 19 f) Hot Standby (see 19.8)

20 **19.1.2 Propagation delay measurement over HDE links**

21 The measurement of propagation delay on an HDE link using the peer-to-peer delay mechanism is

22 illustrated in Figure 11-1 and is described in 11.1.2, with the following differences:

- 23 a) The peer delay initiator is restricted to each timeReceiver port
- 24 b) The timeTransmitter port does not initiate the peer-to-peer delay mechanism.

25 Therefore, pdelayReqSendDisabled and pdelayRespSendDisabled are set as follows:

- 26 c) pdelayReqSendDisabled is set to TRUE for a timeTransmitter port or a PassivePort.
- 27 d) pdelayRespSendDisabled is set to TRUE for a timeReceiver port or a PassivePort.

28 Other configuration values for item c) and item d) are not intended for use with HDE links by this standard.

29 **19.1.3 Transport of time-synchronization information**

30 HDE links uses two-step time transport as specified in clause 11.1.3. The variables oneStepReceive (see

31 19.2.13.9), oneStepTransmit (see 19.2.13.10), and oneStepTxOper (see 19.2.13.11) are set to FALSE for

32 HDE links. Other configuration values for oneStepTransmit and oneStepTxOper are not intended for use

33 with HDE links by this standard.

34 NOTE—The transport of time-synchronization information by a PTP Instance, using Sync and Follow_Up messages, is

35 illustrated in Figure 11-2.

1 19.1.4 Model of operation

2 The model for a PTP Instance of a time-aware system with full-duplex point-to-point links is shown in
3 Figure 11-3. This clause reuses Figure 11-3 (as its structure is unchanged for this clause), where all
4 references to clause 11 in Figure 11-3 are to be replaced by references to clause 19 (this clause). The
5 presence of one HDE MD entity per PTP Port is assumed.

6 A general, media-independent description of the generation of timestamps is given in 8.4.3. A more specific
7 description for PTP event messages is given in 11.3.2.1. A PTP event message is timestamped relative to the
8 LocalClock entity when the message timestamp point (see 3.17) crosses the timestamp measurement plane
9 (see 3.33). The timestamp is corrected for any ingressLatency or egressLatency (see 8.4.3) to produce a
10 timestamp relative to the reference plane (see 3.26). The corrected timestamp value is provided to the MD
11 entity.

12 The MD entity behavior and detailed state machines specific to full-duplex point-to-point links, which are
13 described in 11.2, are reused for HDE links subject to the conditions defined in 19.2.

14 19.2 State machines for HDE links

15 19.2.1 General

16 The state machines for HDE links are described in 11.2.1.

17 19.2.2 Determination of asCapable and asCapableAcrossDomains

18 As specified in 11.2.2.

19 NOTE—For full-duplex point-to-point links, asCapableAcrossDomains is one global variable for all domains per
20 linkport. For HDE links, asCapableAcrossDomains is a global variable per PTP Instance (i.e., per domain), the name
21 asCapableAcrossDomains has been kept for backwards compatibility with existing implementations.

22 19.2.3 Use of MAC Control PAUSE operation

23 As specified in 11.2.3.

24 19.2.4 Use of priority-based flow control

25 As specified in 11.2.4.

26 19.2.5 Use of link aggregation

27 As specified in 11.2.5.

28 19.2.6 Service interface primitives and data structures communicated between state 29 machines

30 The following two subclauses describe the service primitives and data structures communicated between the
31 time-synchronization state machines of the MD entity. First the service primitives are described, followed by
32 the data structures.

33 19.2.7 DL-UNITDATA.request

34 This service primitive is as specified in 11.2.7.

1 **19.2.8 DL-UNITDATA.indication**

2 This service primitive is as specified in 11.2.8.

3 **19.2.9 MDTimestampReceive**

4 This data structure is as specified in 11.2.9.

5 **19.2.10 MDSyncReceive**

6 This data structure is as specified in 10.2.2.2.

7 **19.2.11 MDSyncSend**

8 This structure is as specified in 10.2.2.1.

9 **19.2.12 Overview of MD entity global variables**

10 The overview of MD entity global variables is given in 11.2.12.

11 **19.2.13 MD entity global variables**

12 **19.2.13.1 currentLogPdelayReqInterval:** This variable is as specified in 11.2.13.1.

13 **19.2.13.2 initialLogPdelayReqInterval:** This variable is as specified in 11.2.13.2.

14 **19.2.13.3 pdelayReqInterval:** This variable is as specified in 11.2.13.3.

15 **19.2.13.4 allowedLostResponses:** This variable is as specified in 11.2.13.4.

16 **19.2.13.5 allowedFaults:** This variable is as specified in 11.2.13.5.

17 **19.2.13.6 isMeasuringDelay:** This variable is as specified in 11.2.13.6.

18 **19.2.13.7 meanLinkDelayThresh:** This variable is as specified in 11.2.13.7.

19 **19.2.13.8 syncSequenceId:** This variable is as specified in 11.2.13.8.

20 **19.2.13.9 oneStepReceive:** This variable is as specified in 11.2.13.9.

21 **19.2.13.10 oneStepTransmit:** This variable is as specified in 11.2.13.10.

22 **19.2.13.11 oneStepTxOper:** This variable is as specified in 11.2.13.11.

23 **19.2.13.12 asCapableAcrossDomains:** This variable is as specified in 11.2.13.12.

24 **19.2.13.13 nrrPdelay:** This variable is as specified in 11.2.13.13.

25 **19.2.13.14 nrrSync:** This variable is as specified in 11.2.13.14.

26 **19.2.13.15 nrrCompMethod:** This variable is as specified in 11.2.13.15.

1 **19.2.14 MDSyncReceiveSM state machine**

2 The MDSyncReceiveSM state machine is as specified in 11.2.14.

3 **19.2.15 MDSyncSendSM state machine**

4 The MDSyncSendSM state machine is as specified in 11.2.15.

5 **19.2.16 OneStepTxOperSetting state machine**

6 This state machine is not used for HDE links.

7 **19.2.17 Common Mean Link Delay Service (CMLDS)**

8 CMLDS has not been qualified for use with HDE links.

9 HDE links use the transport-specific peer-to-peer delay mechanism for each domain. Therefore, if the time-
10 aware system implements multiple domains on an HDE link, each domain uses the transport-specific peer-
11 to-peer delay mechanism on that link regardless of its domain number.

12 If multiple TimeTransmitter ports are present on an HDE link, they are in different gPTP domains. CMLDS
13 cannot be used because, in general, the TimeTransmitter ports can be on physical ports of different time-
14 aware systems (i.e., different bridges). In this case, both meanLinkDelay and neighborRateRatio between a
15 PTP End Instance in one of the domains and the TimeTransmitter it is communicating with can be different
16 from meanLinkDelay and neighborRateRatio between a PTP End Instance in another domain on the same
17 end station and the TimeTransmitter that PTP End Instance is communicating with. For instance, in Figure
18 19-1 the meanLinkDelay and neighborRateRatio between the PTP End Instance over domain-m and over
19 domain-n at end station-0 are computed using timestamps from two distinct time aware systems, GM-
20 domain-m and GM-domain-n, respectively; and hence does not use CMLDS.

21 **19.2.18 Common Mean Link Delay Service (CMLDS) global variables**

22 The Common Mean Link Delay Service (CMLDS) global variables are not used for HDE links.

23 **19.2.19 MDPdelayReq state machine**

24 The MDPdelayReq state machine is as specified in 11.2.19.

25 The variable pdelayReqSendDisabled is set per 19.1.2.

26 **19.2.20 MDPdelayResp state machine**

27 The MDPdelayResp state machine is as specified in 11.2.20.

28 The variable pdelayRespSendDisabled is set per 19.1.2.

29 **19.2.21 LinkDelayIntervalSetting state machine**

30 This state machine is not used for HDE links.

31 **19.3 Message attributes**

32 Message attributes is as specified in 11.3.

1 19.4 Message formats

2 Message formats is as specified in 11.4, except for 11.4.2.4, where the domainNumber for Pdelay_Req,
3 Pdelay_Resp, and Pdelay_Resp_Follow_Up messages is the domain number of the HDE gPTP domain used
4 by the transport-specific peer delay mechanism. The domainNumber for all other PTP messages is as
5 specified in 10.6.2.2.6.

6 19.5 Protocol timing characterization

7 19.5.1 General

8 This subclause specifies timing attributes for the media-dependent sublayer specified in this clause.

9 19.5.2 Message transmission intervals

10 19.5.2.1 General interval specification

11 The general message transmission interval specification is as specified in 11.5.2.1.

12 19.5.2.2 Pdelay_Req message transmission interval

13 Pdelay_Req message transmission interval is as specified in 11.5.2.2. The variable
14 useMgtSettableLogPdelayReqInterval is set to TRUE. Other configuration values for
15 useMgtSettableLogPdelayReqInterval are not intended for use with HDE links by this standard.

16 19.5.2.3 Sync message transmission interval default value

17 The Sync message transmission interval default value is as specified in 11.5.2.3.

18 19.5.3 allowedLostResponses

19 The variable allowedLostResponses is as specified in 11.5.3.

20 19.5.4 allowedFaults

21 The variable allowedFaults is as specified in 11.5.4.

22 19.6 Control of computation of neighborRateRatio

23 The control of computation of neighborRateRatio is as specified in 11.6.

24 19.7 Control of computation of meanLinkDelay

25 The control of computation of meanLinkDelay is as specified in 11.7.

26 19.8 HDE links settings and configuration

27 This subclause provides settings and configurations that are specific for HDE links, other configuration
28 values are not intended for use with HDE links by this standard.

- 1 a) The per PTP Instance global variable `externalPortConfigurationEnabled` is set to TRUE for HDE
2 links.
 - 3 b) Both `GtpCapableTransmit` and `GtpCapableReceive` state machines are disabled for HDE links,
4 and therefore `gtpCapableStateMachinesEnabled` are set to FALSE (see 10.4.1). According to
5 10.4.1, if the managed object `gtpCapableStateMachinesEnabled` is FALSE, the global variable
6 `neighborGtpCapable` for the port (see 10.2.5.16) is set to TRUE.
 - 7 c) The variables `useMgtSettableLogSyncInterval` and `useMgtSettableLogAnnounceInterval` are set to
8 TRUE. As a consequence, the `SyncIntervalSetting` and the `AnnounceIntervalSetting` state machines
9 are not used for HDE links.
 - 10 d) The `GtpCapableIntervalSetting` state machine is not used for HDE links. The variable
11 `useMgtSettableLogGtpCapableMessageInterval` is set to TRUE.
- 12 Hot Standby (see Clause 18) has not been qualified for use with HDE links.

Annex A

(normative)

Protocol Implementation Conformance Statement (PICS) proforma³

Add a row at the end of Table A.5 as follows:

A.5 Major Capabilities

Item	Feature	Status	References	Support
MDHDE	Does the PTP Instance support media-dependent HDE link functionality on one or more PTP Ports?	O:1	5.9, 11, 19, A.6, A.22	Yes [] No []

Change A.6 as follows:

A.6 Media access control methods

Item	Feature	Status	References	Support
MAC-IEEE-802.3	Which MAC methods are implemented in conformance with the relevant MAC standards?	O:2	11.1 19.1	Yes [] No [] Yes [] No []
MAC-IEEE-802.11		O:2	12.1	Yes [] No []
MAC-1	Has a PICS been completed for each of the MAC methods implemented as required by the relevant MAC Standards?	M		Yes []
MAC-2	Do all the MAC methods implemented support the MAC Timing aware Service as specified?	M	Clause 11 Clause 12 Clause 13 Clause 19	Yes []

³ Copyright release for PICS proformas: Users of this standard may freely reproduce the PICS proforma in this annex so that it can be used for its intended purpose and may further publish the completed PICS.

Change A.21 as follows:

A.21 External port configuration

Item	Feature	Status	References	Support
	If item EXT is not supported, mark N/A.			NA []
EXT-1	Does the PTP Instance support the specifications for externalPortConfigurationEnabled value of true?	EXT:M	10.3.1 19.8	Yes []
EXT-2	Does the PTP Instance support the PortAnnounceInformationExt state machine?	EXT:M	10.3.14	Yes []
EXT-3	Does the PTP Instance support the PortStateSettingExt state machine?	EXT:M	10.3.15	Yes []

<<Editor's note: Table A.22 is based on Table A.13 for full-duplex point-to-point link. Please, check.>

Insert Table A.22 as follows:

A.22 Media-dependent, HDE link

Item	Feature	Status	References	Support
MDHDE-1	Does this PTP Port implement the functionality of the MDSyncReceiveSM state machine in compliance with the requirements of 19.2.14, 11.2.14 and Figure 11-6?	MDHDE:M	19.2.14, 11.2.14	Yes []
MDHDE-2	Does this PTP Port implement the functionality of the MDSyncSendSM state machine in compliance with the requirements of 19.2.15, 11.2.15 and Figure 11-7?	MIMSTR and MDHDE:M	19.2.15, 11.2.15	Yes []
MDHDE-3	Does this port implement the functionality of the MDPdelayRequest state machine in compliance with the requirements of 19.1.2, 19.2.19, 11.2.19 and Figure 11-9?	MDHDE:M	19.1.2, 19.2.19, 11.2.19	Yes []
MDHDE-4	Does this port implement the functionality of the MDPdelayResponse state machine in compliance with the requirements of 19.1.2, 19.2.20, 11.2.20 and Figure 11-10?	MDHDE:M,	19.2.20 19.1.2 11.2.20	Yes []
MDHDE-5	Does this PTP Port timestamp Sync messages on ingress with respect to the LocalClock in compliance with 19.3, 11.3.2.1 and 11.3.9?	MDHDE:M	19.3, 11.3.2.1	Yes []
MDHDE-6	Does this PTP Port timestamp Sync messages on egress with respect to the LocalClock in compliance with the requirements of 19.3, 11.3.2.1 and 11.3.9?	MIMSTR and MDHDE:M	19.3 11.3.2.1	Yes []
MDHDE-7	Does this port timestamp Pdelay_Req messages on ingress and egress with respect to the LocalClock in compliance with the requirements of 19.3, 11.3.2.1 and 11.3.9?	MDHDE:M	19.3, 11.3.2.1	Yes []

A.22 Media-dependent, HDE link (continued)

Item	Feature	Status	References	Support
MDHDE-8	Does this port timestamp Pdelay_Resp messages on ingress and egress with respect to the LocalClock in compliance with the requirements of 19.3, 11.3.2.1 and 11.3.9?	MDHDE:M	19.3, 11.3.2.1	Yes []
MDHDE-9	Are all IEEE 802.1AS messages on this port sent without a Q-tag in compliance with the requirements of 19.3, 11.3.3?	MDHDE:M	19.3, 11.3.3	Yes []
MDHDE-10	Do all media-dependent messages transmitted on this port use a destination MAC address taken from Table 11-3 in compliance with the requirements of 19.3 and 11.3.4 [01-80-C2-00-00-0E]?	MDHDE:M	19.3, 11.3.4	Yes []
MDHDE-11	Do all media-dependent messages transmitted on this port use a source MAC address that is assigned to that port in compliance with the requirements of 19.3 and 11.3.4?	MDHDE:M	19.3, 11.3.4	Yes []
MDHDE-12	Do all media-dependent message transmitted on this port use an EtherType specified in Table 11-4 [88-F7]?	MDHDE:M	19.3, 11.3.5	Yes []
MDHDE-13	Does the header of all the media-dependent messages on this port comply with the requirements of 19.4, 11.4.2 and Table 10-7?	MDHDE:M	19.4, 11.4.2	Yes [] N/A []
MDHDE-14	Does the body of Sync messages sent on this PTP Port comply with the requirements of 19.4, 11.4.3, Table 11-8, and Table 11-9?	MDHDE:M	19.4, 11.4.3	Yes []
MDHDE-15	Does the body of Follow_Up messages sent on this PTP Port comply with the requirements of 19.4, 11.4.4, 6.4.3.3 (lastGmPhaseChange), and Table 11-10?	MDHDE:M	19.4, 11.4.4, 6.4.3.3	Yes []
MDHDE-16	Does the body of Pdelay_Req messages sent on this port comply with the requirements of 19.4, 11.4.5 and Table 11-12?	MDHDE:M	19.4, 11.4.5	Yes []
MDHDE-17	Does the body of Pdelay_Resp messages sent on this port comply with the requirements of 19.4, 11.4.6 and Table 11-13?	MDHDE:M	19.4, 11.4.6	Yes []
MDHDE-18	Does the body of Pdelay_Resp_Follow_Up messages sent on this port comply with the requirements of 19.4, 11.4.7 and Table 11-14?	MDHDE:M	19.4, 11.4.7	Yes []
MDHDE-19	Are all reserved fields in media-dependent messages sent on this port set to 0 in compliance with the requirements of 19.4, 11.4.1?	MDHDE:M	19.4, 11.4.1	Yes []
MDHDE-20	Do the Sync message sequence numbers comply with the requirements of 19.3, 11.3.8?	MIMSTR and MDHDE:M	19.3, 11.3.8	Yes [] N/A []

A.22 Media-dependent, HDE link (continued)

Item	Feature	Status	References	Support
MDHDE-21	Do the Pdelay_Req message sequence numbers comply with the requirements of 19.3 and 11.3.8?	MDHDE:M	19.3, 11.3.8	Yes []
MDHDE-22	Does the Pdelay mean request transmission interval comply with the requirements of 19.5.2.2 and 11.5.2.2?	MDHDE:M	19.5.2.2, 11.5.2.2	Yes []
MDHDE-23	Does the Sync mean transmission interval comply with the requirements of 19.5.2.3 and 11.5.2.3?	MDHDE:M	19.5.2.3, 11.5.2.3	Yes []
MDHDE-24	Does the HDE media-dependent layer set the asCapable global variable in the media-independent PortSync entity in compliance with the requirements of 19.2.2 and 11.2.2?	MDHDE:M	19.2.2, 11.2.2	Yes []
MDHDE-25	Does the PTP Instance consider the PTP Port or Link Port, respectively, to not be exchanging Pdelay messages when a valid response is not received in compliance with the requirements of 19.5.3 and 11.5.3?	MDHDE:M	19.5.3, 11.5.3	Yes []
MDHDE-26	Does the PTP Instance ignore TLVs, of PTP messages, that it cannot parse and attempt to parse the next TLV, in compliance with the requirements of 19.4 and 11.4.1?	MDHDE:M	19.4, 11.4.1	Yes []
MDHDE-27	Does the time-aware system initialize meanLinkDelayThresh as specified in 19.2.2 and 11.2.2?	MDHDE:M	19.2.2, 11.2.2	Yes []
MDHDE-28	Does this port support propagation delay averaging?	MDHDE:O	19.2.19, 11.2.19.3.4	Yes [] No []
MDHDE-29	Does this port support two-step capability on receive?	MDHDE:M	19.2.14, 11.2.14, item d) of 5.5	Yes []
MDHDE-30	Does this port support two-step capability on transmit?	MDHDE:M	19.2.15, 11.2.15, item e) of 5.5	Yes []
MDHDE-31	Is the transport-specific peer-to-peer delay mechanism supported?	MDHDE:M	19.2.17	Yes []

Annex F

(informative)

PTP profile included in this standard

F.1 General

Change the paragraph in F.1 as follows:

The specification in this standard of synchronized time transport over a full-duplex point-to-point link includes a PTP profile. The information contained in a PTP profile is described in 20.3 of IEEE Std 1588-2019. This annex summarizes the PTP profile for transport of timing over full-duplex point-to-point links. This PTP profile is also used [in the transport of timing over HDE links \(see Clause 19\)](#), and in the transport of timing over CSN when a CSN clock reference is not present (see Clause 16). This PTP profile is not used in the transport of timing over IEEE [Std 802.11](#) links and IEEE [Std 802.3](#) EPON links; both these transports use native timing mechanisms to assist in the synchronized time transport.

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